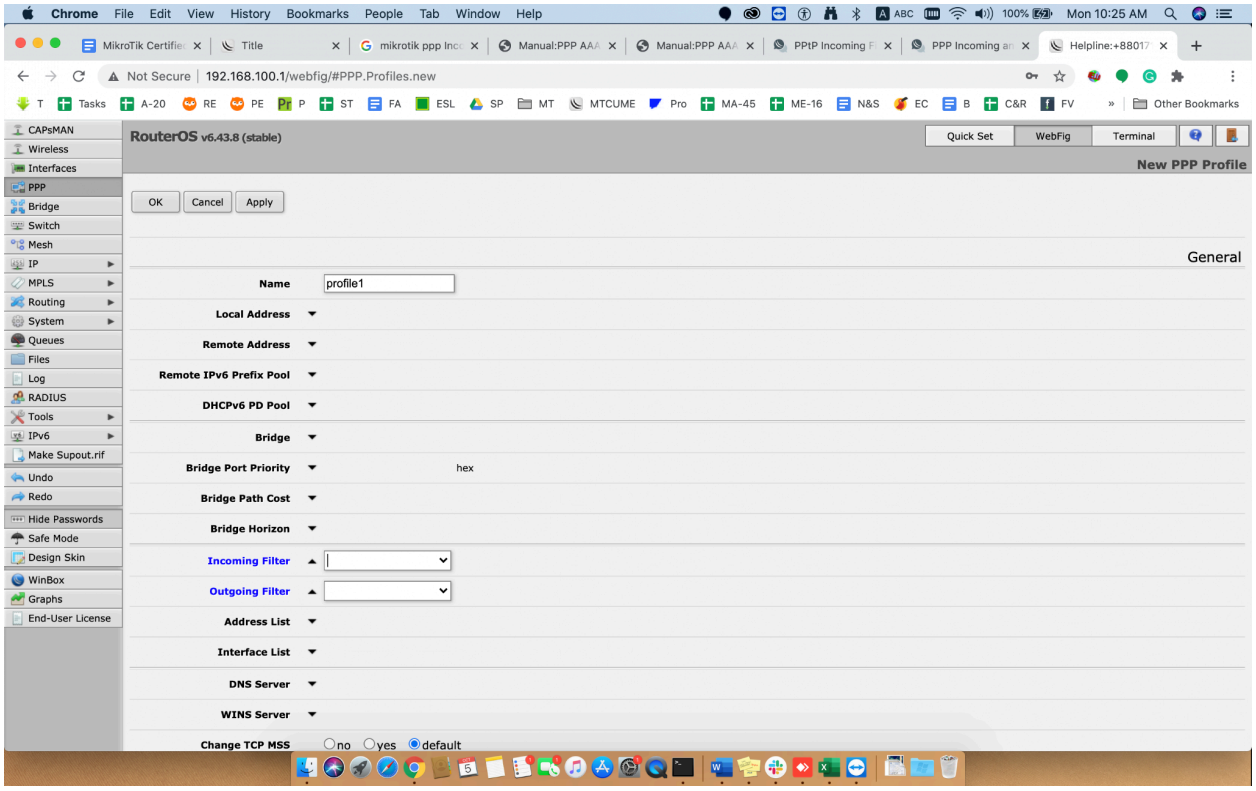
MikroTik Certified User Management Engineer(MTCUME)

PPP
PPP Profile
Sub-menu: /ppp profile
PPP profiles are used to define default values for user access records stored under /ppp secret submenu. Settings in /ppp secret User Database override corresponding /ppp profile settings except that single IP addresses always take precedence over IP pools when specified as local-address or remote-address parameters.

Local and remote addresses
Local address: Tunnel address or name of the pool from which address is assigned to ppp interface locally.

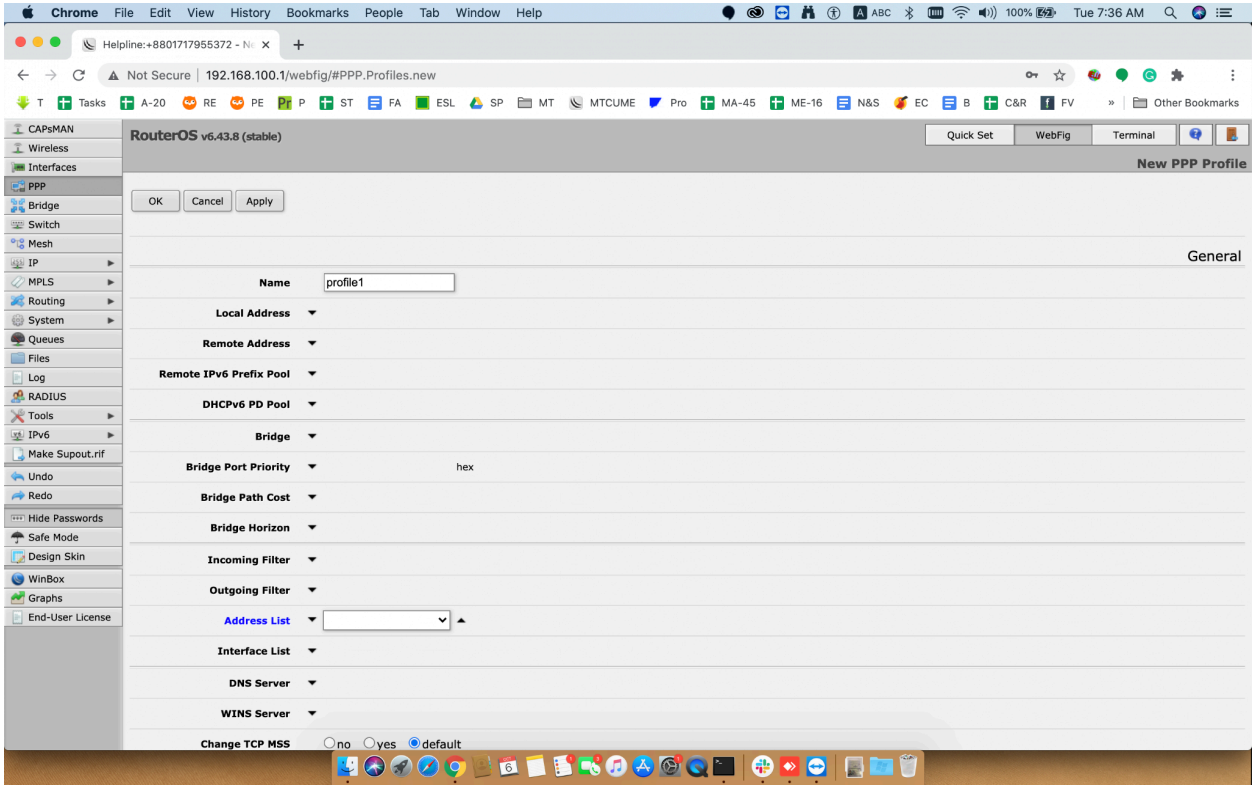
Remote address: Tunnel address or name of the pool from which address is assigned to remote ppp interface.

Incoming and outgoing filters
Incoming-filter
Firewall chain name for incoming packets. Specified chain gets control for each packet coming from the client. The ppp chain should be manually added and rules with action=jump jump-target=ppp should be added to other relevant chains in order for this feature to work.



Outgoing-filter
Firewall chain name for outgoing packets. Specified chain gets control for each packet going to the client. The ppp chain should be manually added and rules with action=jump jump-target=ppp should be added to other relevant chains in order for this feature to work.

Address List
Firewall address lists allow a user to create lists of IP addresses grouped together under a common name. Firewall filter, mangle and NAT facilities can then use those address lists to match packets against them.



Change-tcp-mss
change Maximum Segment Size(MSS)
Modifies connection MSS settings (applies only for IPv4):
yes - adjust connection MSS value
no - do not adjust connection MSS value
default - derive this value from the interface default profile; same as no if this is the interface default profile

Use encryption
use-encryption (yes | no | default | require; Default: default)

Specifies whether to use data encryption or not.
yes - enable data encryption
no - disable data encryption
default - derive this value from the interface default profile; same as no if this is the interface default profile
require - explicitly requires encryption
This setting does not work on OVPN and SSTP tunnels.

session-timeout
Maximum time the connection can stay up. By default no time limit is set.

Rate-limit
Rate limitation in form of rx-rate[/tx-rate] [rx-burst-rate[/tx-burst-rate]

Only-one setting
Defines whether a user is allowed to have more than one ppp session at a time
yes - a user is not allowed to have more than one ppp session at a time
no - the user is allowed to have more than one ppp session at a time
default - derive this value from the interface default profile; same as no if this is the interface default profile

PPP Secret
Service: Specifies the services that particular user will be able to use. service (any | async | isdn | l2tp | pppoe | pptp | ovpn | sstp; Default: any)

Profile: Which user profile to use.

Local-address: IP address that will be set locally on ppp interface.

remote-address:IP address that will be assigned to remote ppp interface.

Routes configuration: Routes that appear on the server when the client is connected.

Limit-bytes-in: Maximal amount of bytes for a session that client can upload.

Limit-bytes-out: Maximal amount of bytes for a session that client can download.

IP Pool
Set addresses ranges: IP address list of non-overlapping IP address.For example, 10.0.0.1-10.0.0.27,10.0.0.32-10.0.0.47

Next pool options: when address is acquired from pool that has no free addresses, and next-pool property is set to another pool, then next IP address will be acquired from next-pool

PPTP, L2TP
Theory
PPTP-Point-to-Point Tunneling Protocol
L2TP-Layer Two Tunneling Protocol
Comparison

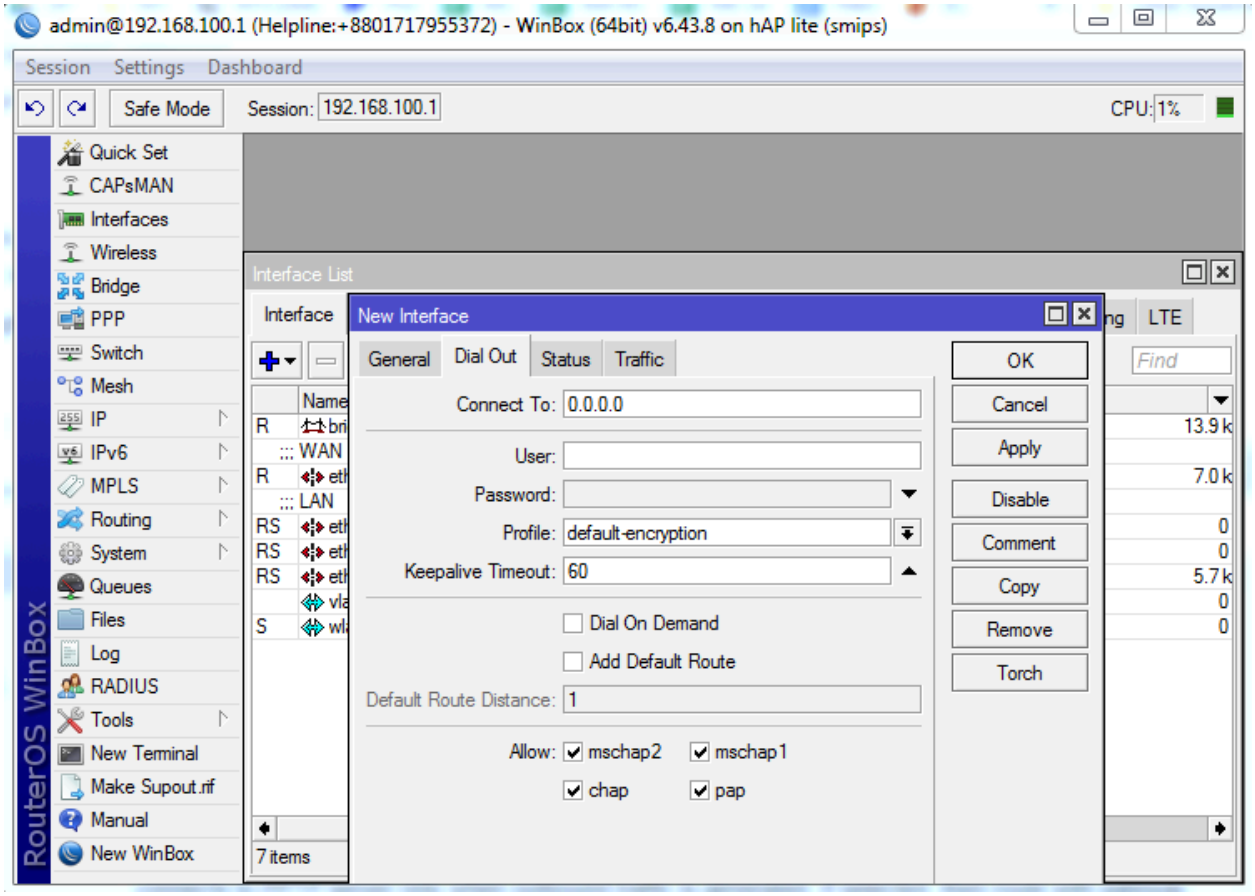
Difference between PPTP and L2TP

Here is a quick difference between the two protocols based on multiple factors:

Factors	PPTP	L2TP
Supported Devices	Compatible with all devices i.e. Windows, Mac, Linux, Android, iOS, Routers, etc	Also compatible with different devices like Windows, Mac, Android, Linux, iOS, Routers.
Encryption	128-bit encryption	256-bit encryption
Stability	Pretty unstable when compared with L2TP, but works with Wi-Fi hotspots	Very much stable and works on every device, networks and platform.
Speed	Faster speed	Slower speed
Usage	It can be used with an in-built client, username and password, and server address	It is used on a VPN client, username and password, server address, and pre-shared key.
Security	Not much secure because it has basic encryption	Very much secure because it completely encrypts your data with strong encryption.

PPTP Client configuration
/interface pptp-client

Client setup

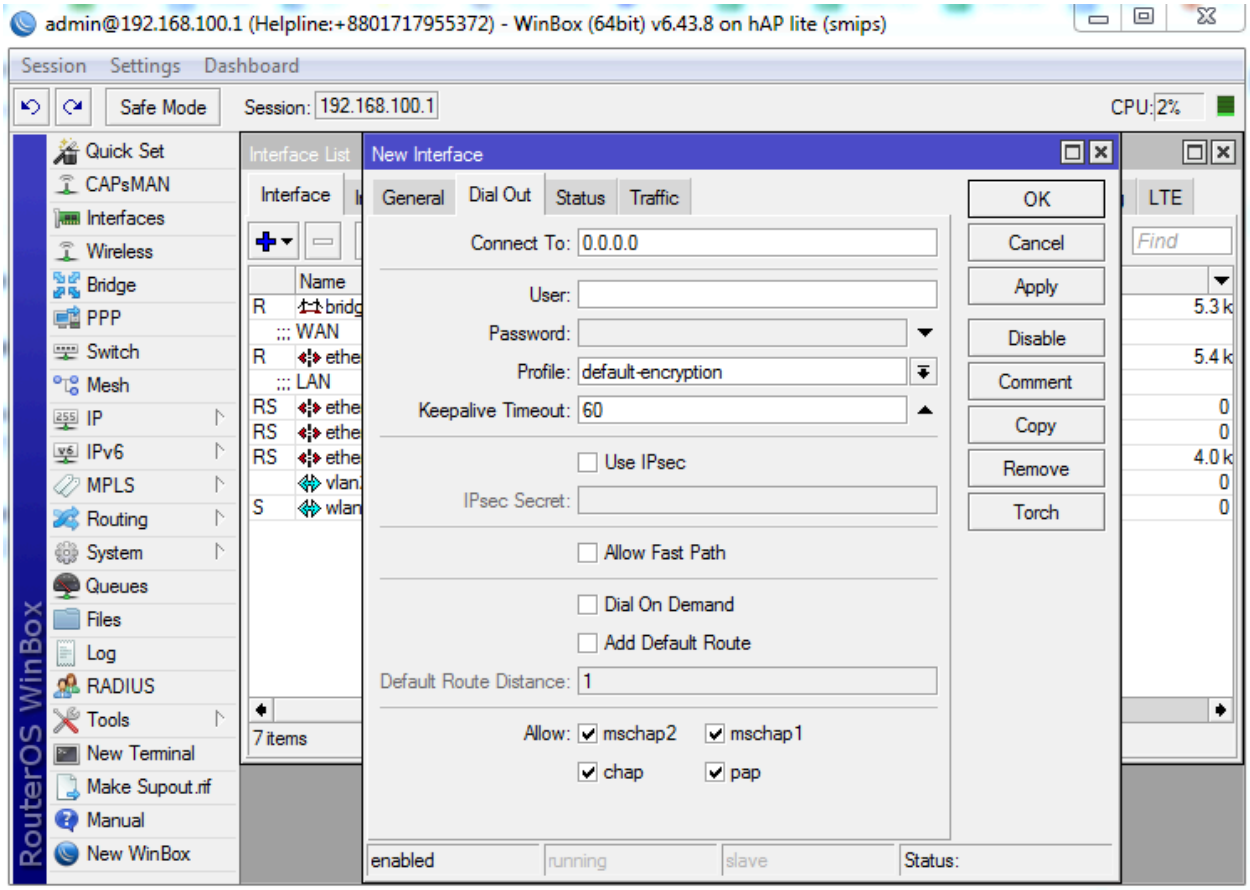


Set profile: Bandwidth profile

Dial on demand: connects to PPTP server only when outbound traffic is generated.

PPTP Server configuration
Check PPTP lecture sheet

L2TP Client configuration
/interface l2tp-client



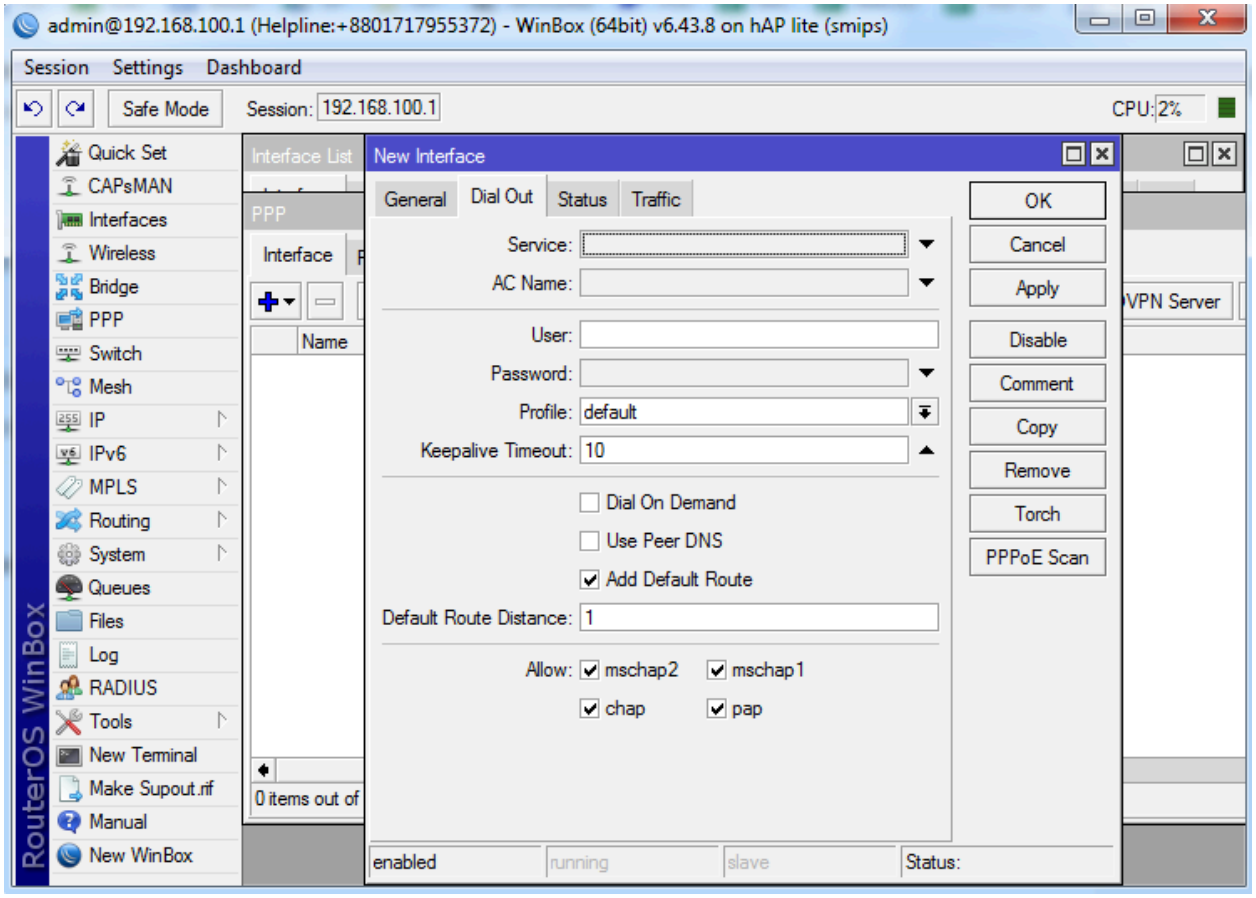
L2TP Server configuration
Check l2tp lecture sheet

PPPoE server and client
Point-to-Point Protocol over Ethernet

Comparison to other PPP protocols

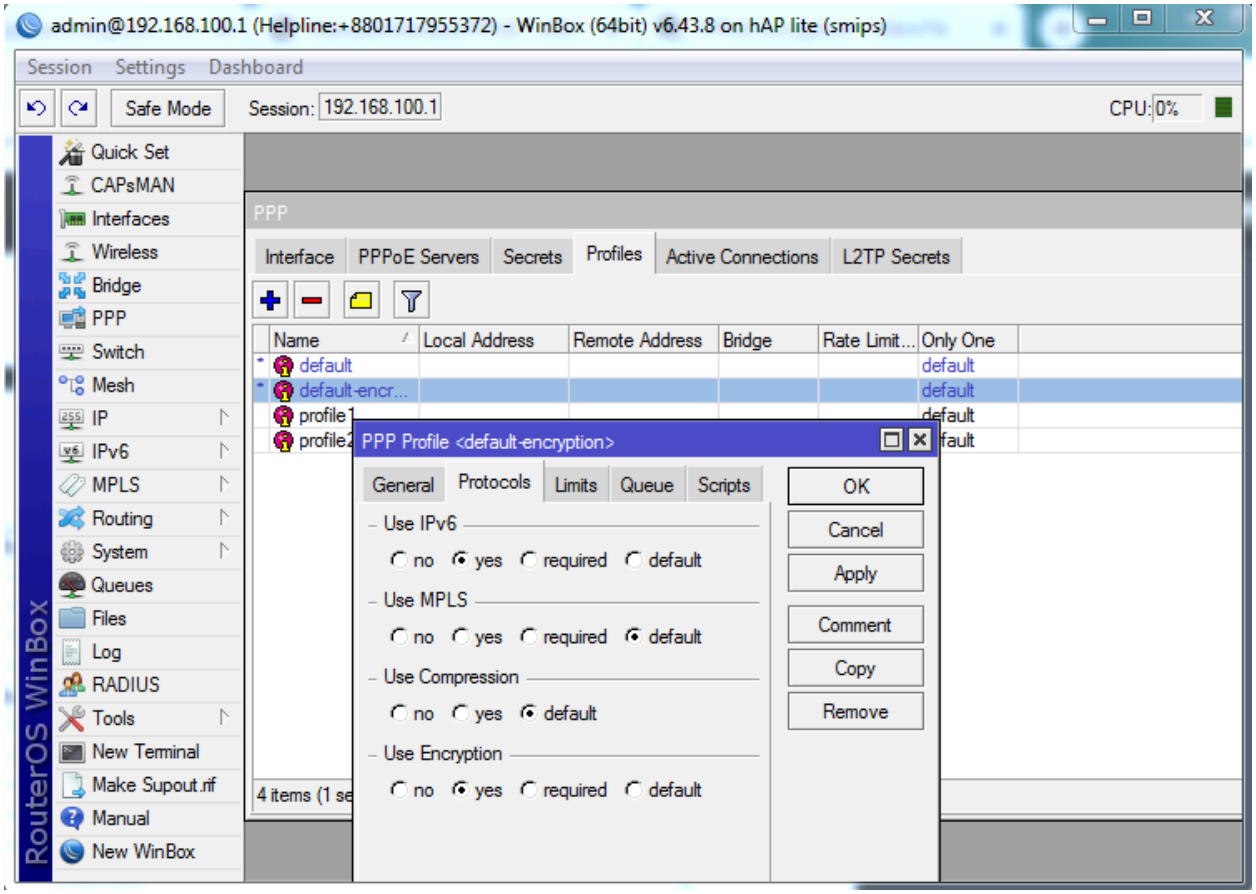
BASIS FOR COMPARISON	HDLC	PPP
Expands to	High-level Data Link Layer Protocol	Point-to-Point Protocol
Type of protocols	Bit-oriented protocol	Byte oriented protocol
Used in	Only synchronous media	Synchronous as well as asynchronous media
Authentication	No provision of authentication	Provides authentication
Dynamic addressing	Does not offer dynamic addressing.	Dynamic addressing is used.
Implemented in	Point-to-point and multipoint configurations.	Only point-to-point configurations.
Compatibility with other protocols	Can not be operated with non-Cisco devices.	Interoperable with non-Cisco devices also.

PPPoE client configuration

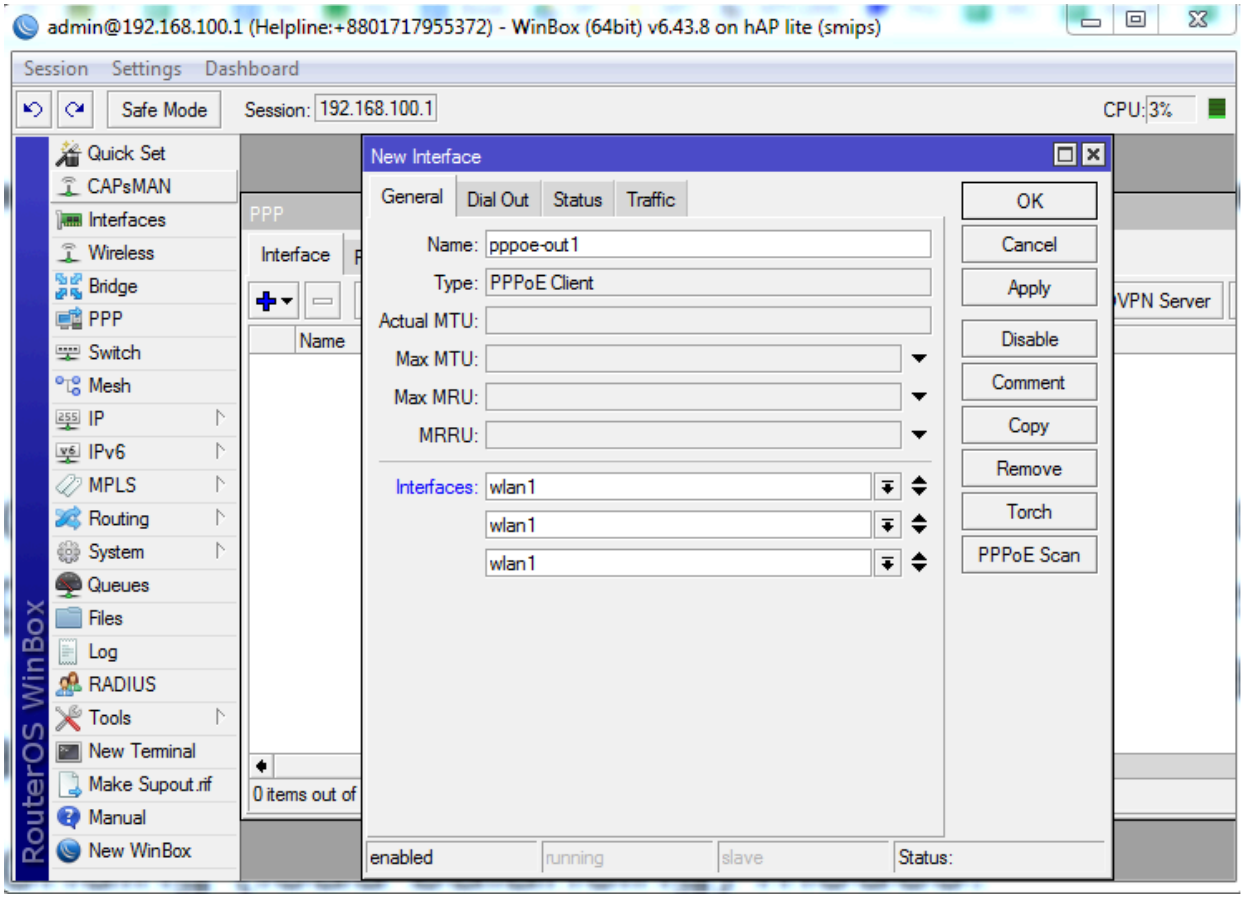


PPPoE Server configuration
Check PPPOE lecture sheet

Encryption



Interface ECMP



Bridging

Set L2TP tunnel

Check l2tp tunnel lecture sheet

If you wish traffic for the remote office to go over tunnel you will need to add a specific static route as follows:

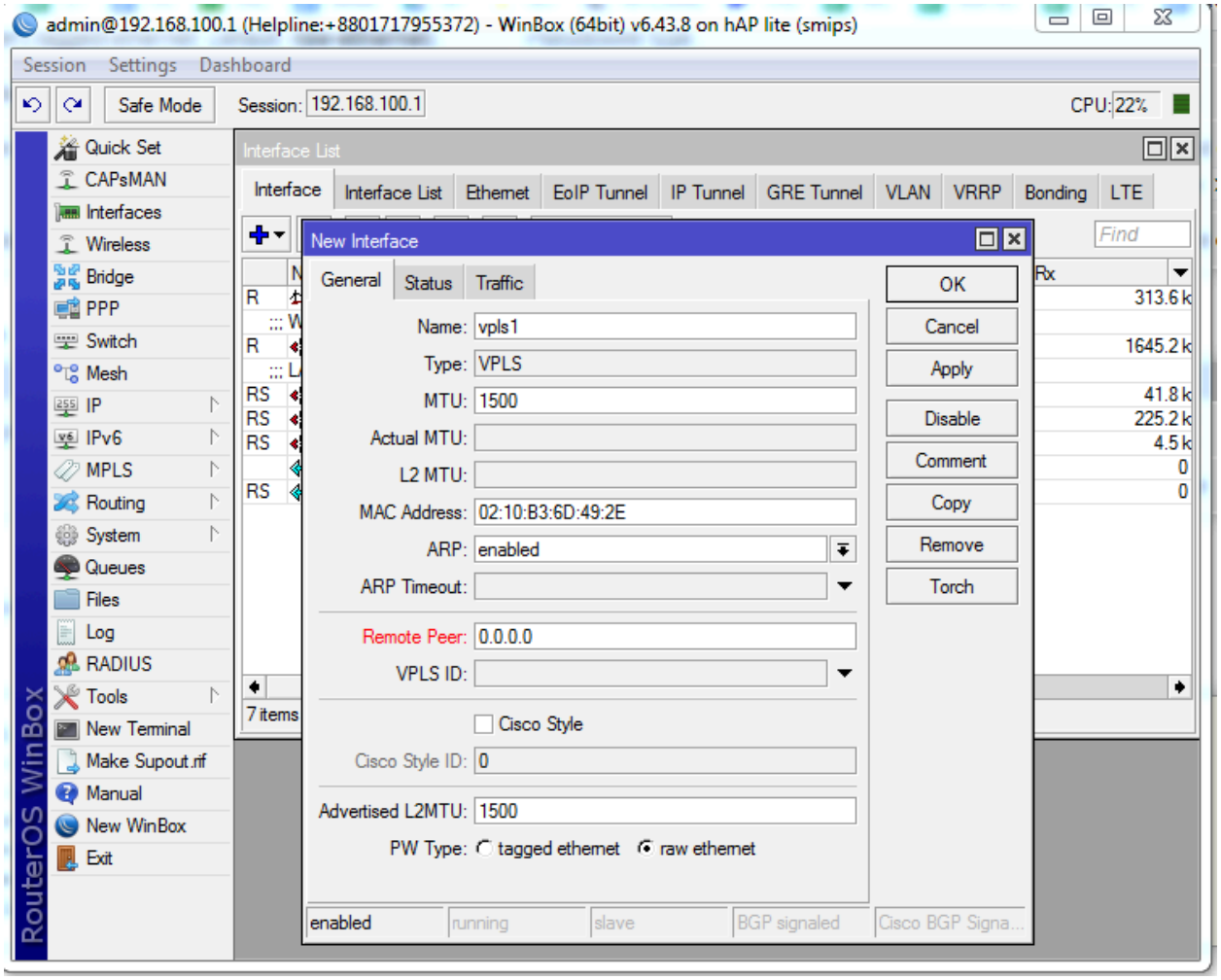
ip route> add dst-address=10.1.101.0/24 gateway=l2tp-out1

Set EoIP tunnel

Check EOIP tunnel lecture sheet

Set VPLS tunnel

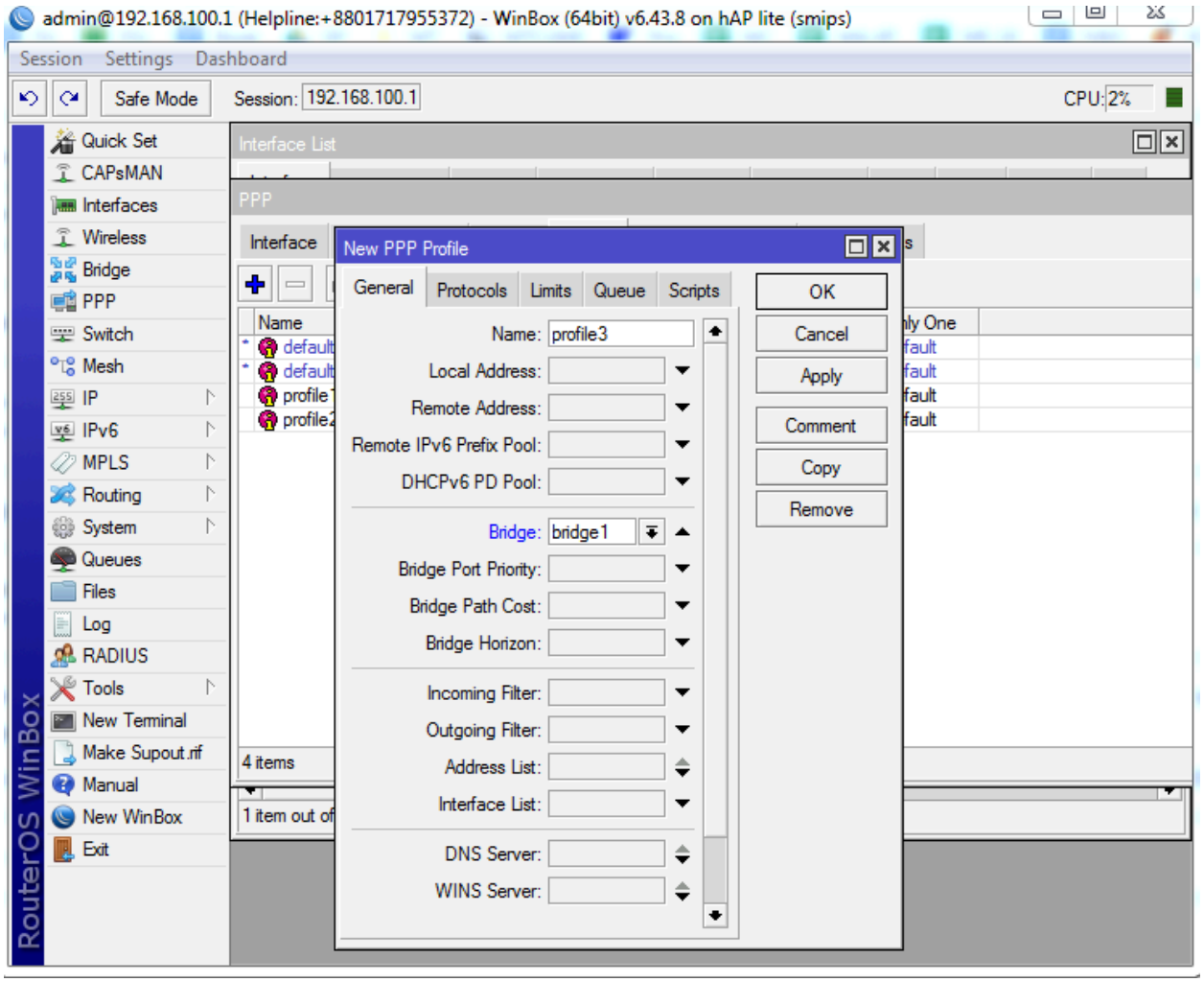
Virtual Private Lan Service (VPLS) interface can be considered tunnel interface just like EoIP interface. To achieve transparent ethernet segment forwarding between customer sites.



BCP

BCP is Bridge Control Protocol, allows sending Ethernet Frame over PPP.

BCP (Bridge Control Protocol) should be enabled on both sides (PPP server and PPP client) to make it work.We need to interconnect two remote offices and make them in one Ethernet network. We have requirement to use encryption to protect data exchange between two offices

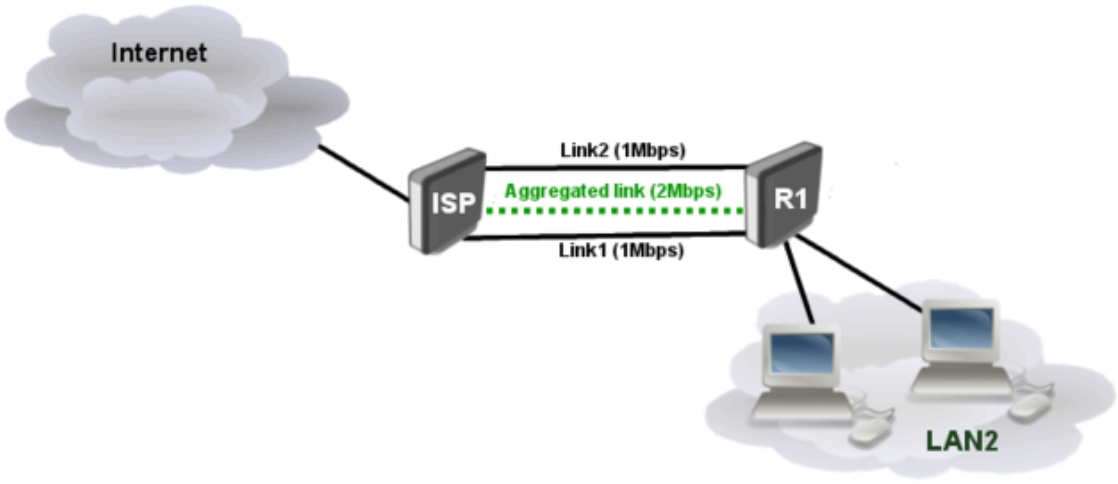


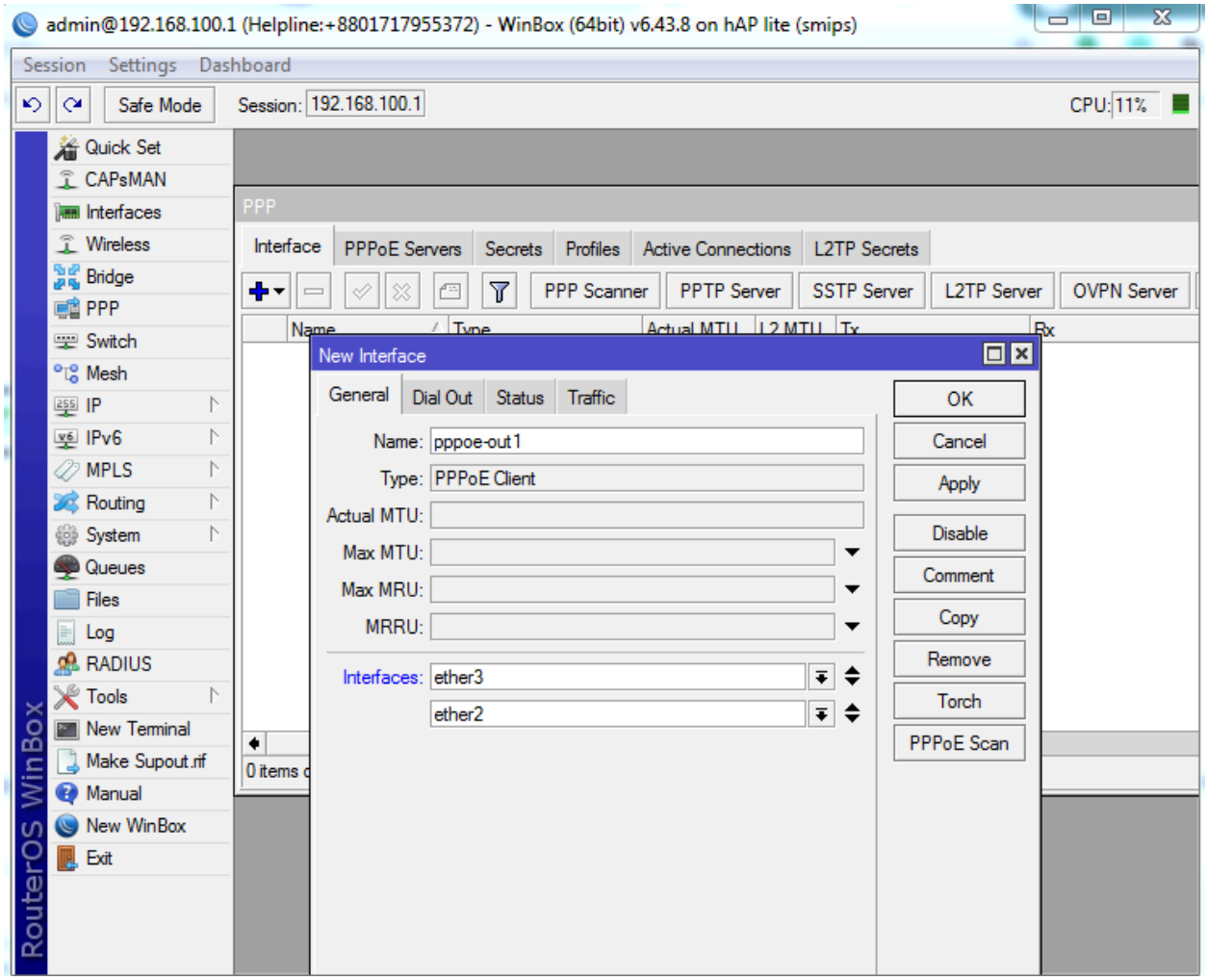
L2TP and BCP

Check the EoIP lecture sheet. Layer 2 tunnel protocol

Multilink Protocol

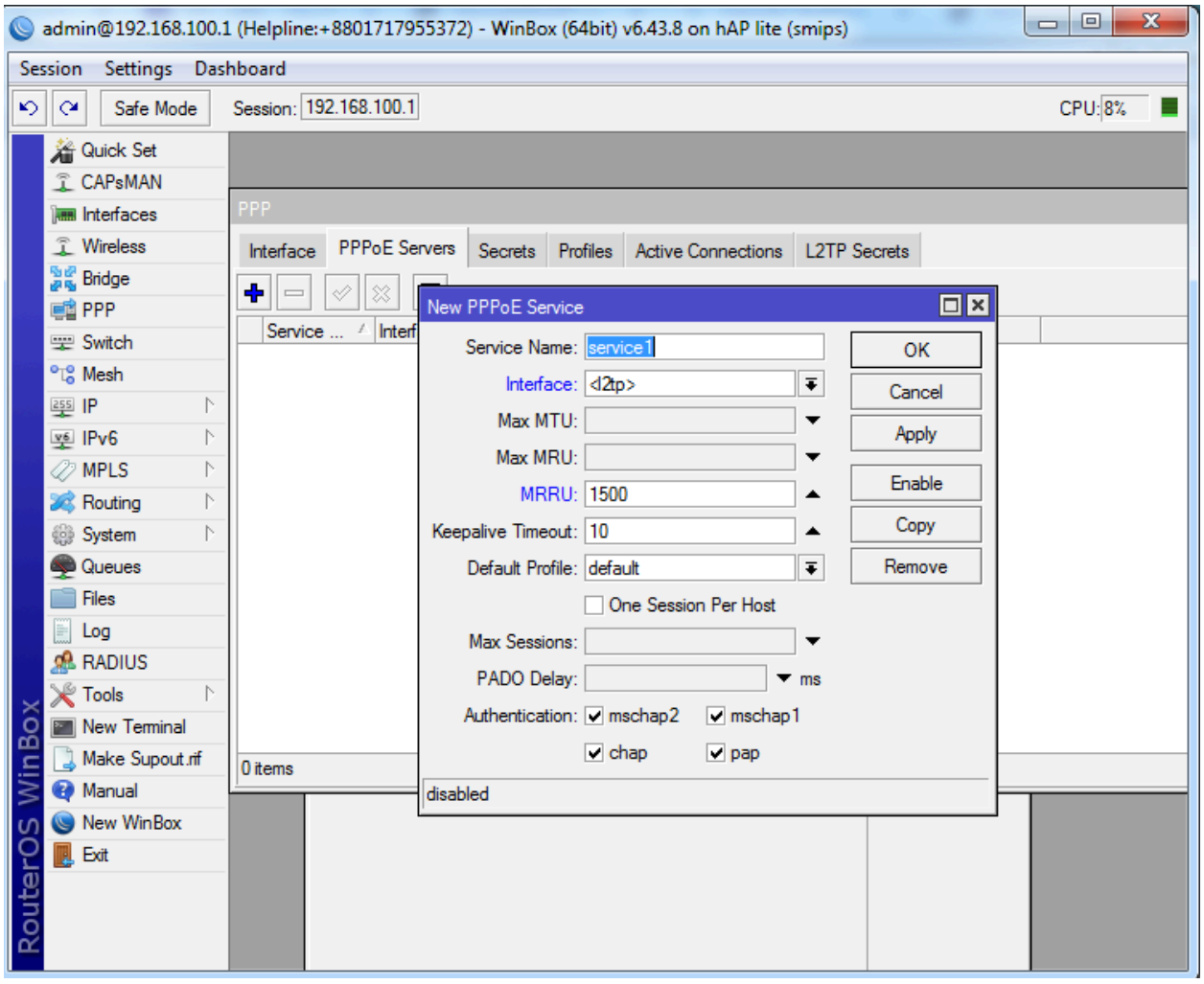
Multi-Link Point to Point Protocol (MP, Multi-Link PPP, MultiPPP or MLPPP) is a method of splitting, recombining, and sequencing data across multiple logical data links.


Internet
Link2 (1Mbps)
ISP
Aggregated link (2Mbps)
R1
Link1 (1Mbps)
LAN2


admin@192.168.100.1 (Helpline:+8801717955372) - WinBox (64bit) v6.43.8 on hAP lite (smips)
Session Settings Dashboard
Safe Mode
Session: 192.168.100.1
CPU: 11%
Quick Set
CAPsMAN
Interfaces
Wireless
Bridge
PPP
Switch
Mesh
IP
IPv6
MPLS
Routing
System
Queues
Files
Log
RADIUS
Tools
New Terminal
Make Supout.rif
Manual
New WinBox
Exit
RouterOS WinBox
PPP
Interface
PPPoE Servers
Secrets
Profiles
Active Connections
L2TP Secrets
PPP Scanner
PPTP Server
SSTP Server
L2TP Server
OVPN Server
Name
Type
Actual MTU
L2 MTU
Tx
Rx
0 items
New Interface
General
Dial Out
Status
Traffic
Name: pppoe-out1
Type: PPPoE Client
Actual MTU:
Max MTU:
Max MRU:
MRRU:
Interfaces: ether3
ether2
OK
Cancel
Apply
Disable
Comment
Copy
Remove
Torch
PPPoE Scan

Enable multilink by specifying correct MRRU settings
To make it work the Multilink Protocol uses additional LCP configuration options Multilink Maximum Received Reconstructed Unit (MRRU)

To enable Multi-link PPP over single link you must specify MRRU (Maximum Receive Reconstructed Unit) option. If both sides support this feature there are no need for MSS adjustment (in firewall mangle). Study shows that MRRU is less CPU expensive that 2 mangle rules per client. MRRU allows to divide packet to multiple channels therefore increasing possible MTU and MRU (up to 65535 bytes)



Set PPPoE server with MLPPP support
Let's configure pppoe server compatible with Windows clients and MRRU enabled.

/interface pppoe-server server> add service-name=myPPP interface=ether1 mrru=1614
In short - standard PPP link - just specify MRRU in both sides :)
Presently there are no MLPPP server support available.

IPsec
IPSec stands for IP Security. Internet Protocol Security (IPsec) is a set of protocols defined by the Internet Engineering Task Force (IETF) to secure packet exchange over unprotected IP/IPv6 networks such as Internet.
Comparison to other VPN protocols

	OpenVPN	PPTP	L2TP/IPsec	SoftEther	WireGuard	SSTP	IKEv2/IP
Encryption	160-bit, 256-bit	128-bit	256-bit	256-bit	ChaCha20	256-bit	256-bit
Security	Very high	Weak	High security (might be weakened by NSA)	High	High	High	High
Speed	Fast	Speedy, due to low encryption	Medium, due to double encapsulation	Very fast	Fast	Fast	Very fast
Stability	Very stable	Very stable	Stable	Very stable	Not yet stable	Very stable	Very stal
Compatibility	Strong desktop support, but mobile could be improved. Requires third-party software.	Strong Windows desktop support.	Multiple device and platform support.	Multiple desktop and mobile OS support. No native operating system support.	Linux, being built for other platforms and operating systems.	Windows-platform, but works on other Linux distributions.	Limited platform support beyond Window Blackber
Final Word	Most recommended choice. Fast and secure.	Native on Windows. Weak security. Useful for geo-restricted content.	Versatile and secure. A decent alternative to OpenVPN.	Up and coming. Flexible, fast, and secure. A great alternative to OpenVPN.	Has promise to be fast and efficient. Still in development.	Faster and more secure alternative to PPTP and L2TP.	Secure, stable, a mobile-oriented

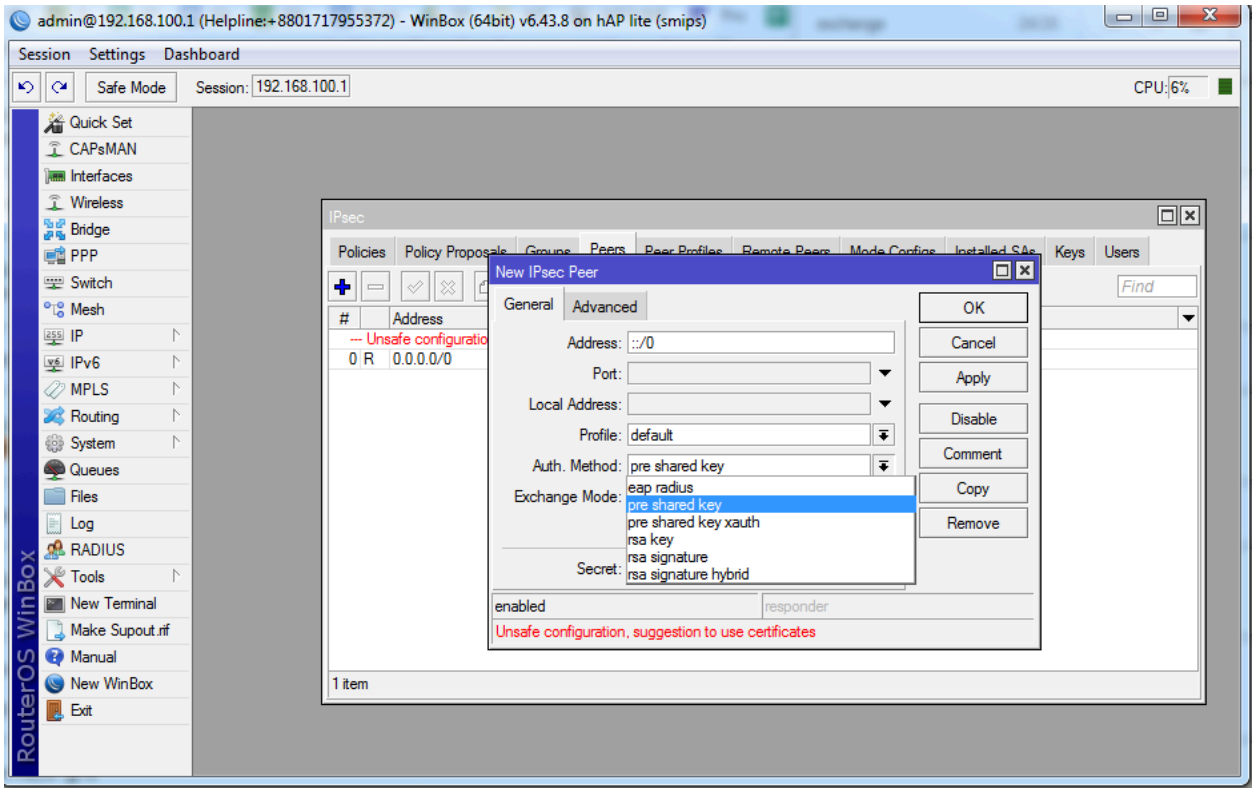
ipsec Peers
Sub-menu: /ip ipsec peer

Peer configuration settings are used to establish connections between IKE daemons. This connection then will be used to negotiate keys and algorithms for SAs. Exchange mode is the only unique identifier between the peers, meaning that there can be multiple peer configurations with the same remote-address as long as different exchange-mode is used.

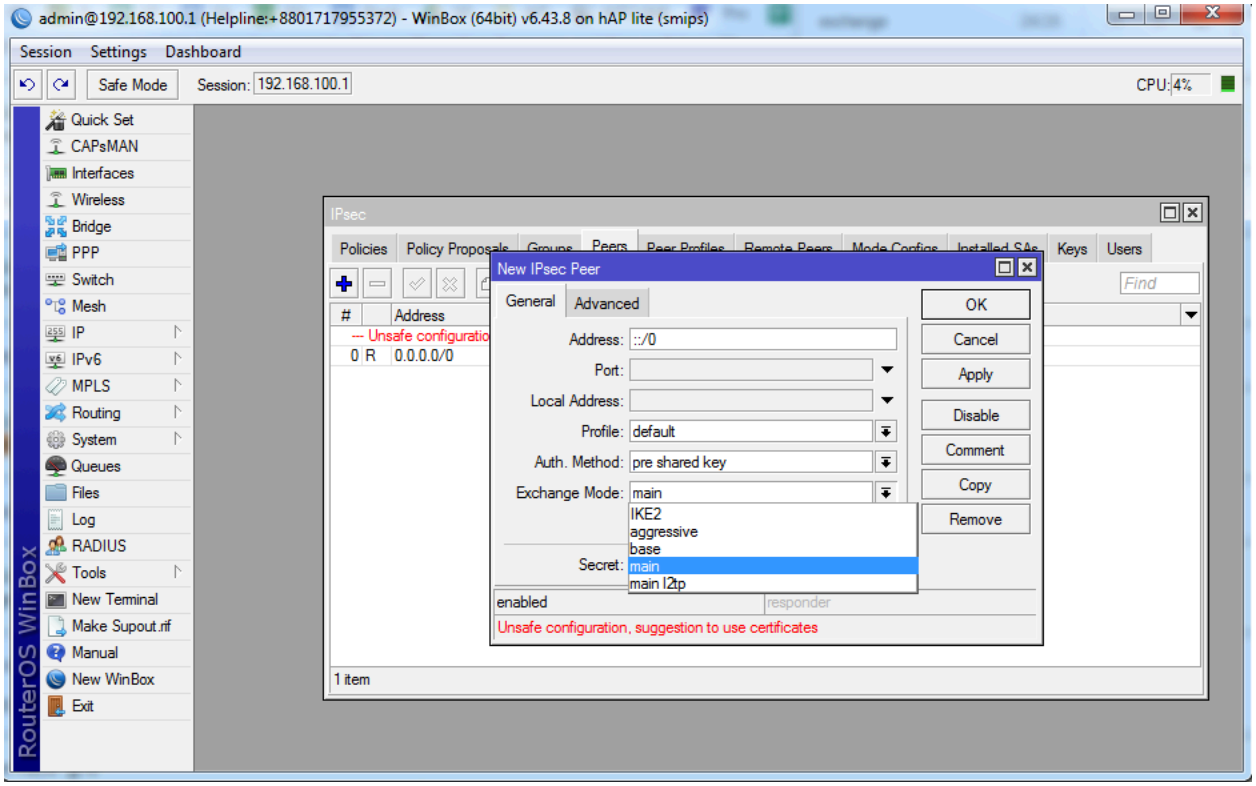
Use different authentication methods

Authentication method:
digital-signature - authenticate using a pair of RSA certificates;

eap - IKEv2 EAP authentication for initiator (peer with netmask of /32). Must be used together with eap-methods;
eap-radius - IKEv2 EAP RADIUS passthrough authentication for responder (RFC 3579). Server certificate in this case is required. If server certificate is not specified then only clients supporting EAP-only (RFC 5998) will be able to connect. Note that EAP method should be compatible with EAP-only;
pre-shared-key - authenticate by a password (pre-shared secret) string shared between the peers (not recommended since offline attack on pre-shared key is possible);
rsa-key - authenticate using a RSA key imported in keys menu. Only supported in IKEv1;
pre-shared-key-xauth - authenticate by a password (pre-shared secret) string shared between the peers + XAuth username and password. Only supported in IKEv1;
rsa-signature-hybrid - responder certificate authentication with initiator XAuth. Only supported in IKEv1.



IPsec exchange modes



Different ISAKMP phase 1 exchange modes according to RFC 2408. main mode relaxes rfc2409 section 5.4, to allow pre-shared-key authentication in main mode. ike2 mode enables Ikev2 RFC 7296. Parameters that are ignored by IKEv2 proposal-check, compatibility-options, lifebytes, dpd-maximum-failures, nat-traversal.

Encryption and hash algorithms
RouterOS ESP supports various encryption and authentication algorithms.

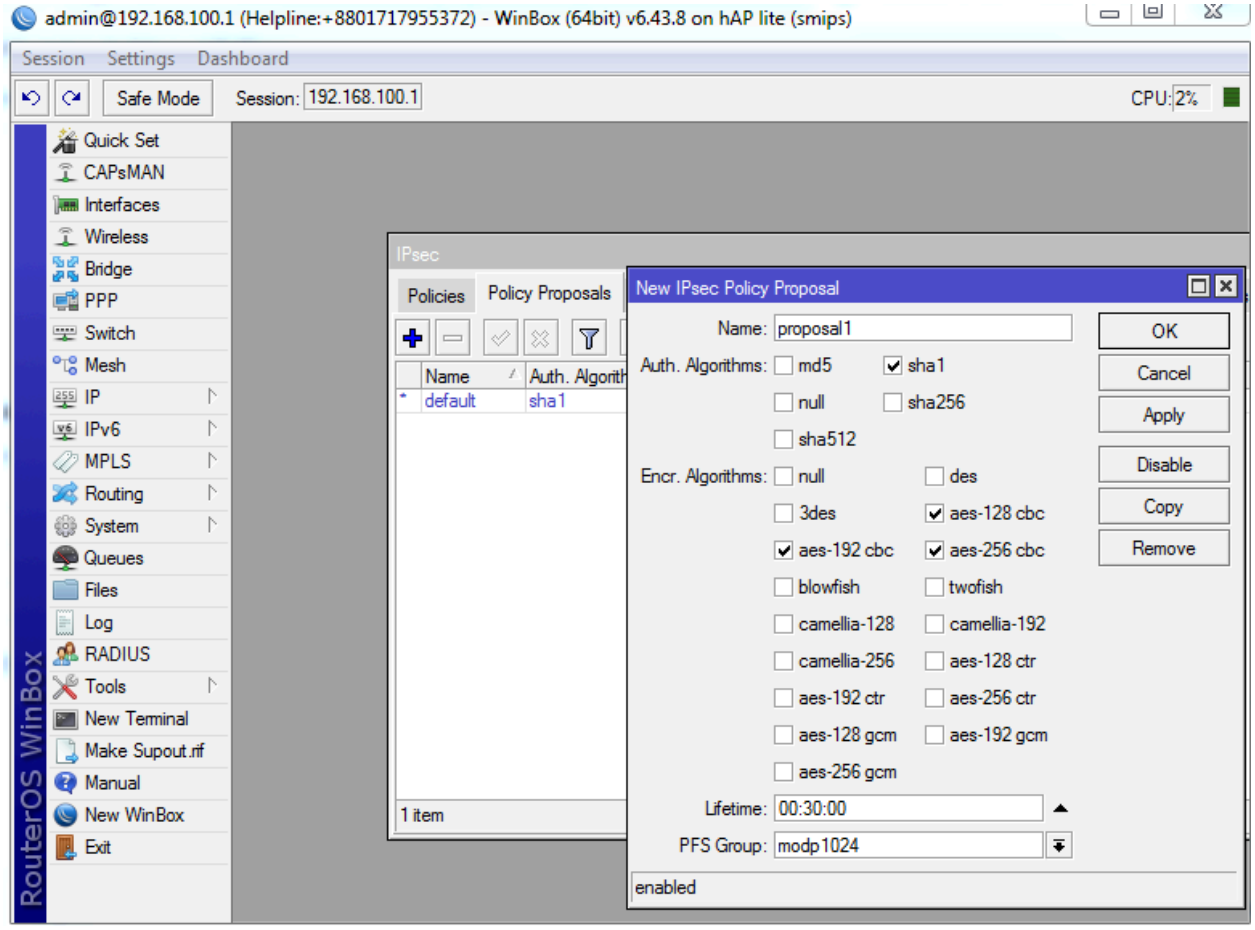
Authentication:

MD5
SHA1
SHA2 (256-bit, 512-bit)
Encryption:

AES - 128-bit, 192-bit and 256-bit key AES-CBC, AES-CTR and AES-GCM algorithms;
Blowfish - added since v4.5
Twofish - added since v4.5
Camellia - 128-bit, 192-bit and 256-bit key Camellia encryption algorithm added since v4.5
DES - 56-bit DES-CBC encryption algorithm;
3DES - 168-bit DES encryption algorithm;

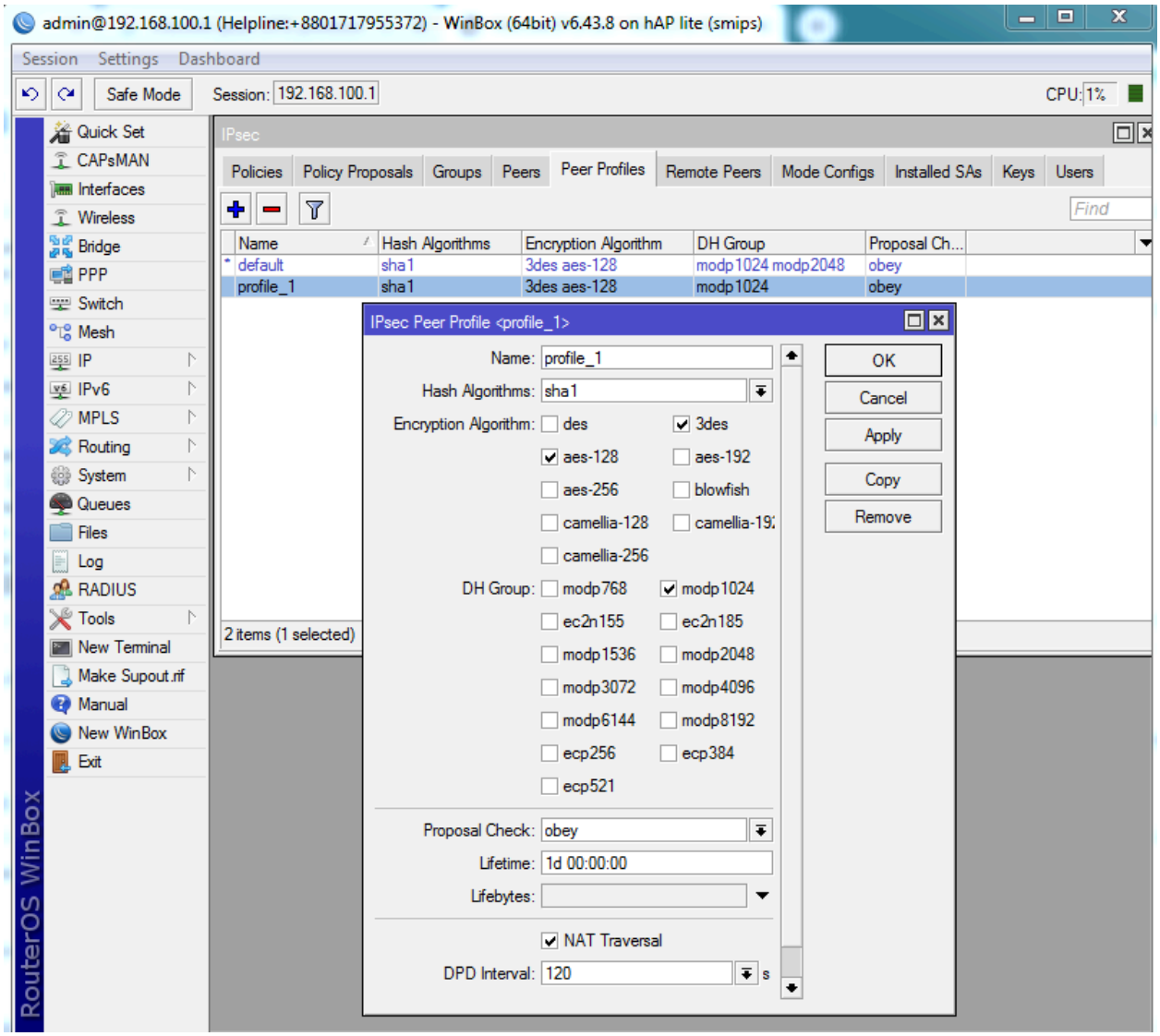
nat-traversal
Use Linux NAT-T mechanism to solve IPsec incompatibility with NAT routers inbetween IPsec peers. This can only be used with ESP protocol (AH is not supported by design, as it signs the complete packet, including IP header, which is changed by NAT, rendering AH signature invalid). The method encapsulates IPsec ESP traffic into UDP streams in order to overcome some minor issues that made ESP incompatible with NAT

lifetime
How long to use SA before throwing it out.



Lifebytes
Phase 1 lifebytes is used only as administrative value which is added to proposal. Used in cases if remote peer requires specific lifebytes value to establish phase 1

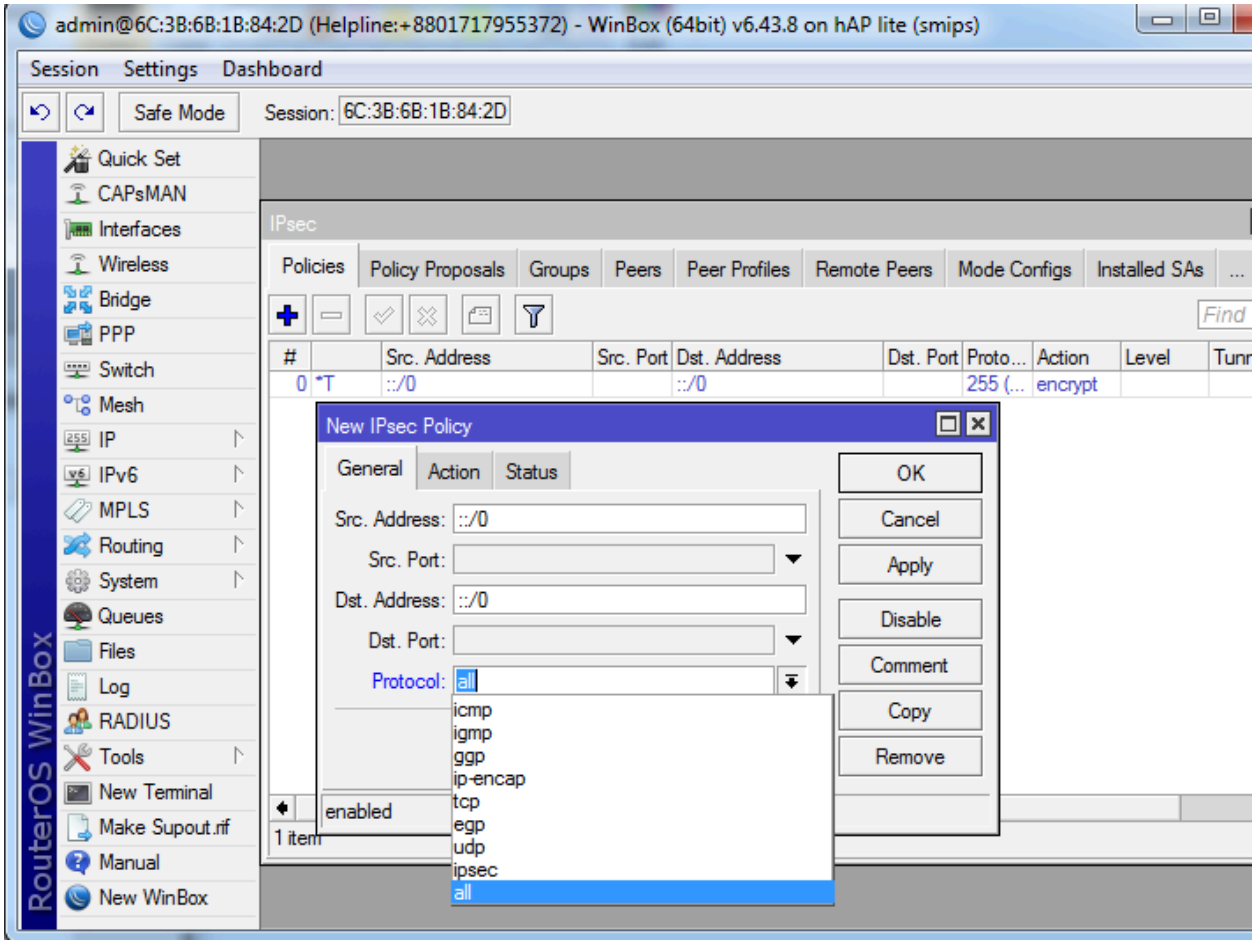
DPD protocol
Dead peer detection interval. If set to disable-dpd, dead peer detection will not be used.
Warning: Phase 1 is not re-keyed if DPD is disabled when lifetime expires, only phase 2 is re-keyed. To force phase 1 re-key, enable DPD.



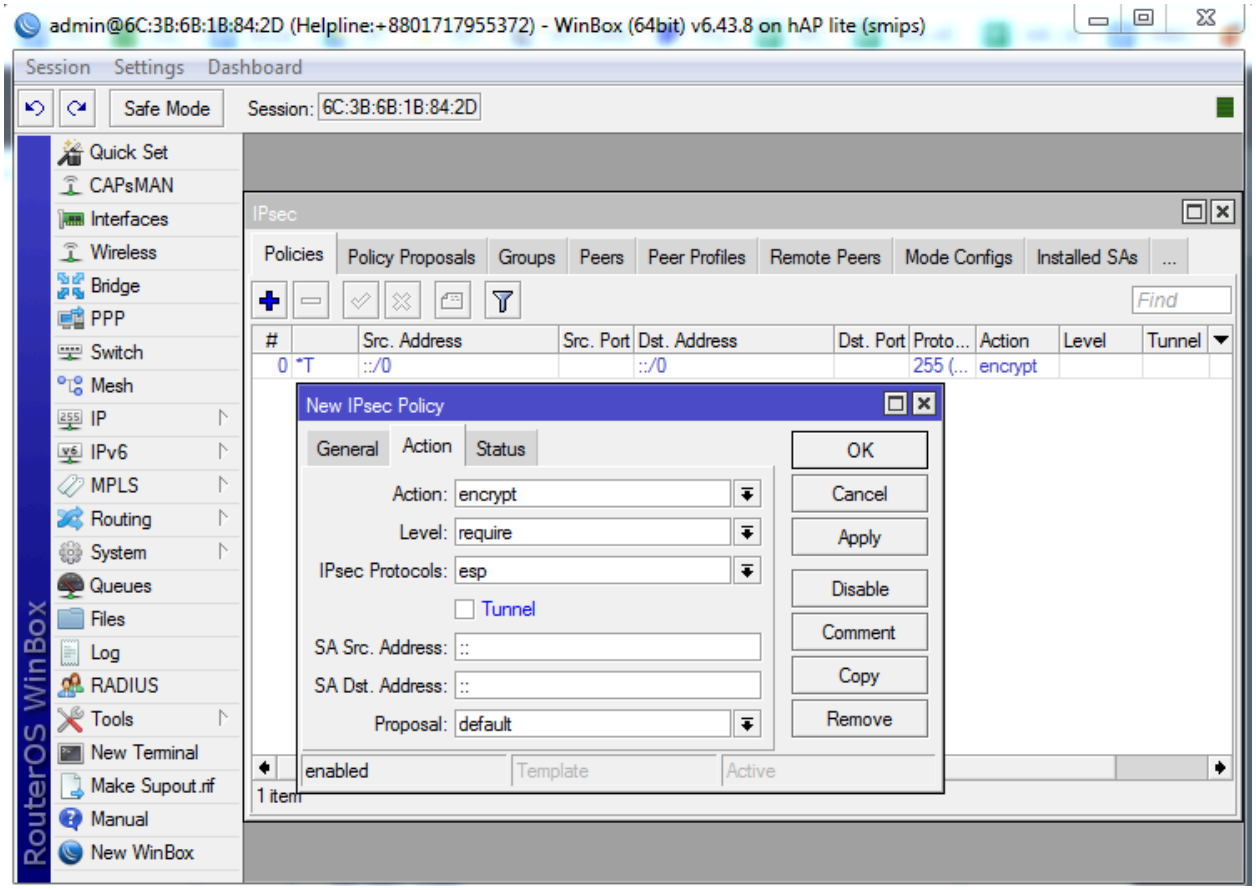
Policy

IPsec protocol and action

IP packet protocol to match.protocol (all | egp | ggp| icmp | igmp | ...; Default: all)



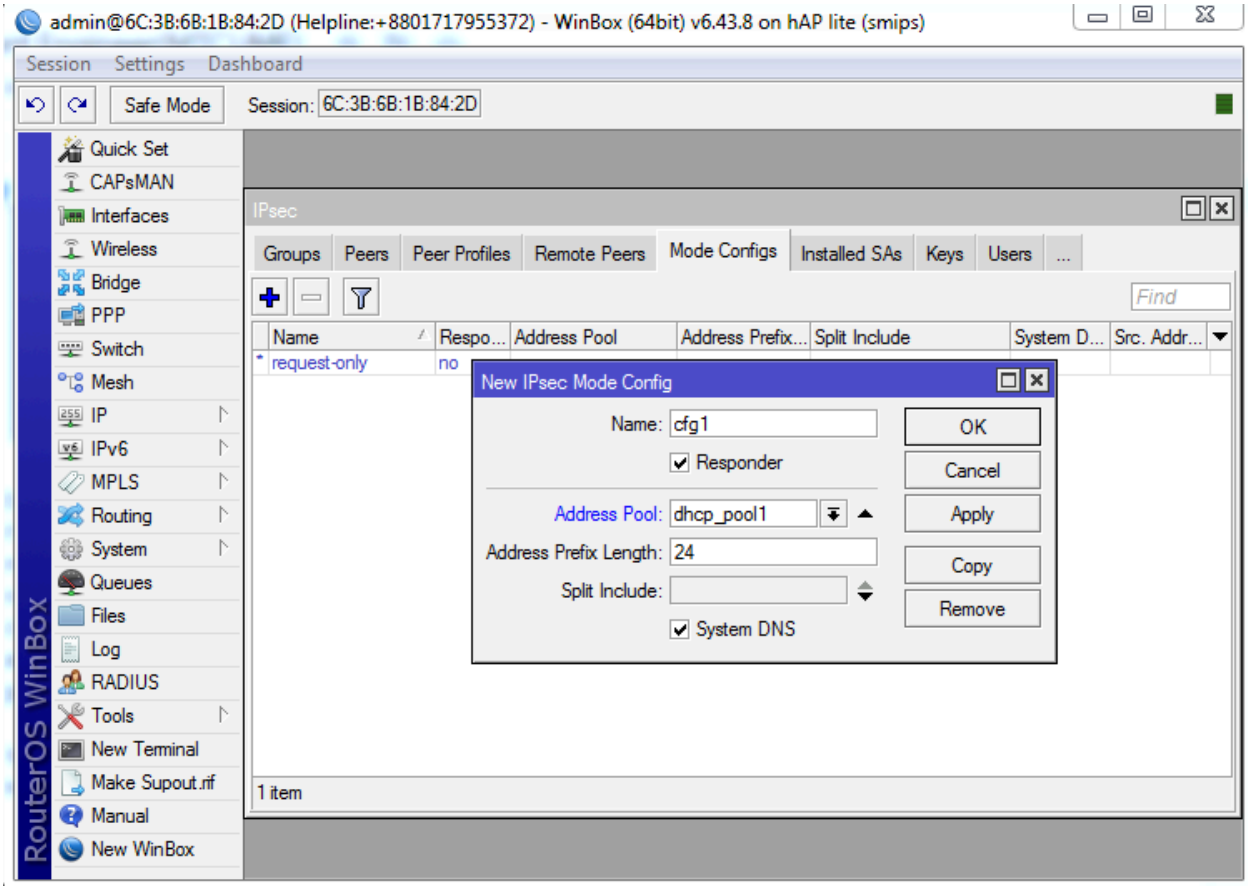
Tunnels
Specifies whether to use tunnel mode.



Enabling dynamic source NAT rule generation

If we look at the generated dynamic policies, we see that only traffic with a specific (received by mode config) source address will be sent through the tunnel. But a router in most cases will need to route a specific device or network through the tunnel. In such case we can use source NAT to change the source address of packets to match the mode config address. Since the mode config address is dynamic, it is impossible to create static source NAT rule. In RouterOS it is possible to generate dynamic source NAT rules for mode config clients.

/ip ipsec mode-config
set [find name=ike2-rw] src-address-list=local



Proposal
/ip ipsec proposal
Proposal information that will be sent by IKE(Internet Key Exchange Protocol) daemons to establish SAs for certain policy.

Encryption and authentication algorithms
Encryption:
AES - 128-bit, 192-bit and 256-bit key AES-CBC, AES-CTR and AES-GCM algorithms;
Blowfish - added since v4.5
Twofish - added since v4.5
Camellia - 128-bit, 192-bit and 256-bit key Camellia encryption algorithm added since v4.5
DES - 56-bit DES-CBC encryption algorithm;
3DES - 168-bit DES encryption algorithm;

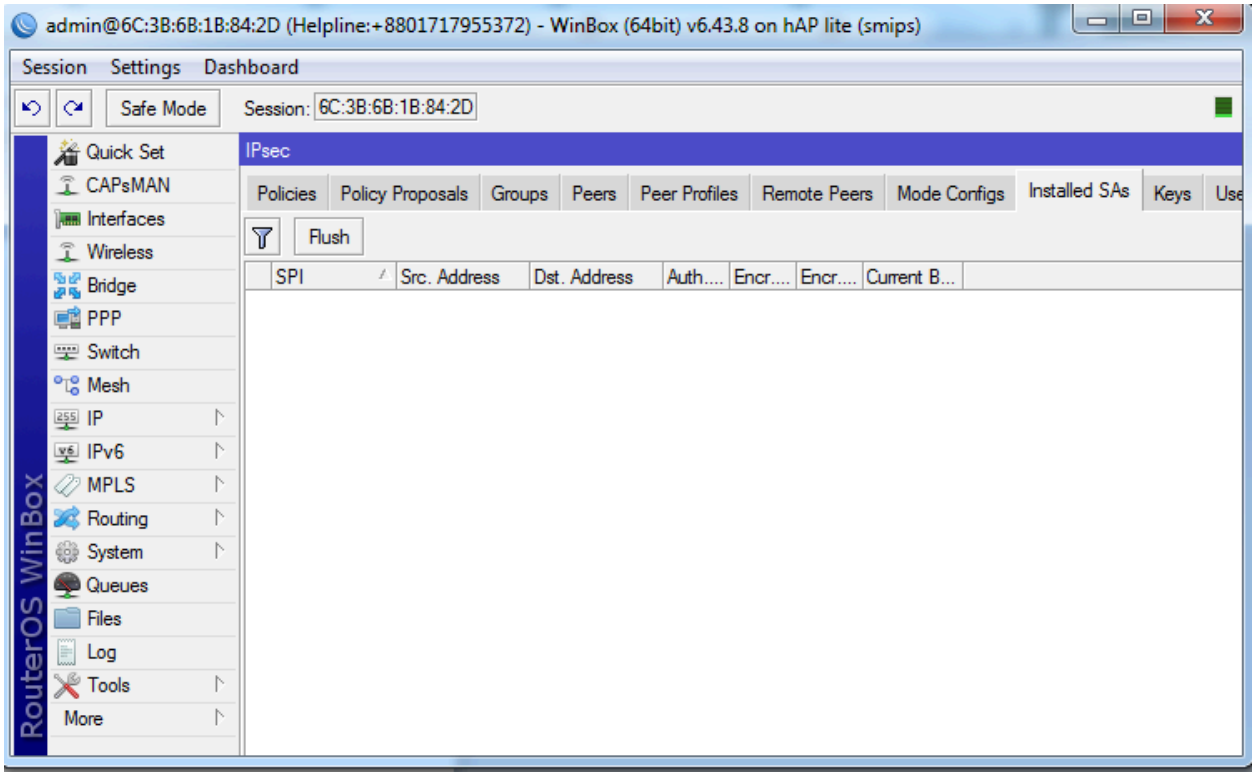
Authentication:
MD5
SHA1
SHA2 (256-bit, 512-bit)

Lifetime
lifetime (time; Default: 30m). How long to use SA before throwing it out.

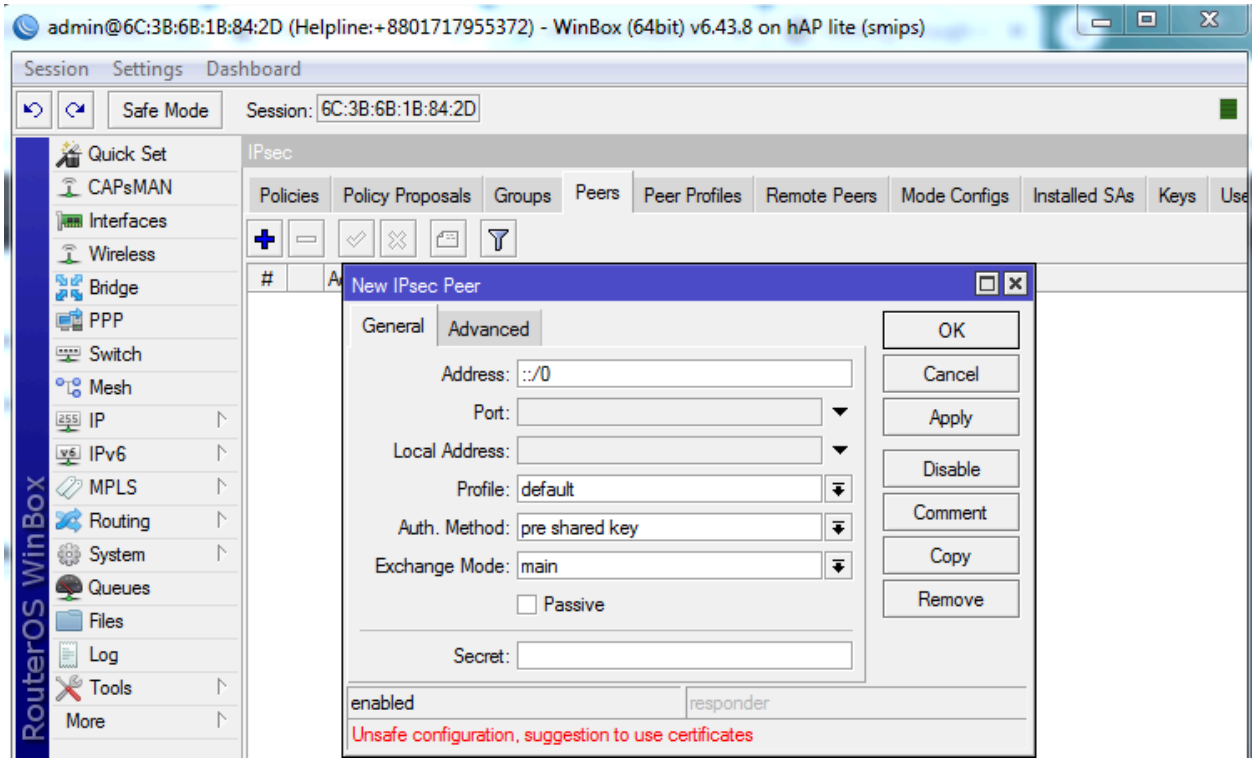
PFS
Perfect Forward Secrecy (PFS)
pfs-group (ec2n155 | ec2n185 | ecp256 | ecp384 | ecp521 | modp768 | modp1024 | modp1536 | modp2048 | modp3072 | modp4096 | modp6144 | modp8192 | none; Default: modp1024)
Diffie-Helman group used for Perfect Forward Secrecy.

Installed-SA
/ip ipsec installed-sa
This menu provides information about installed security associations including the keys.

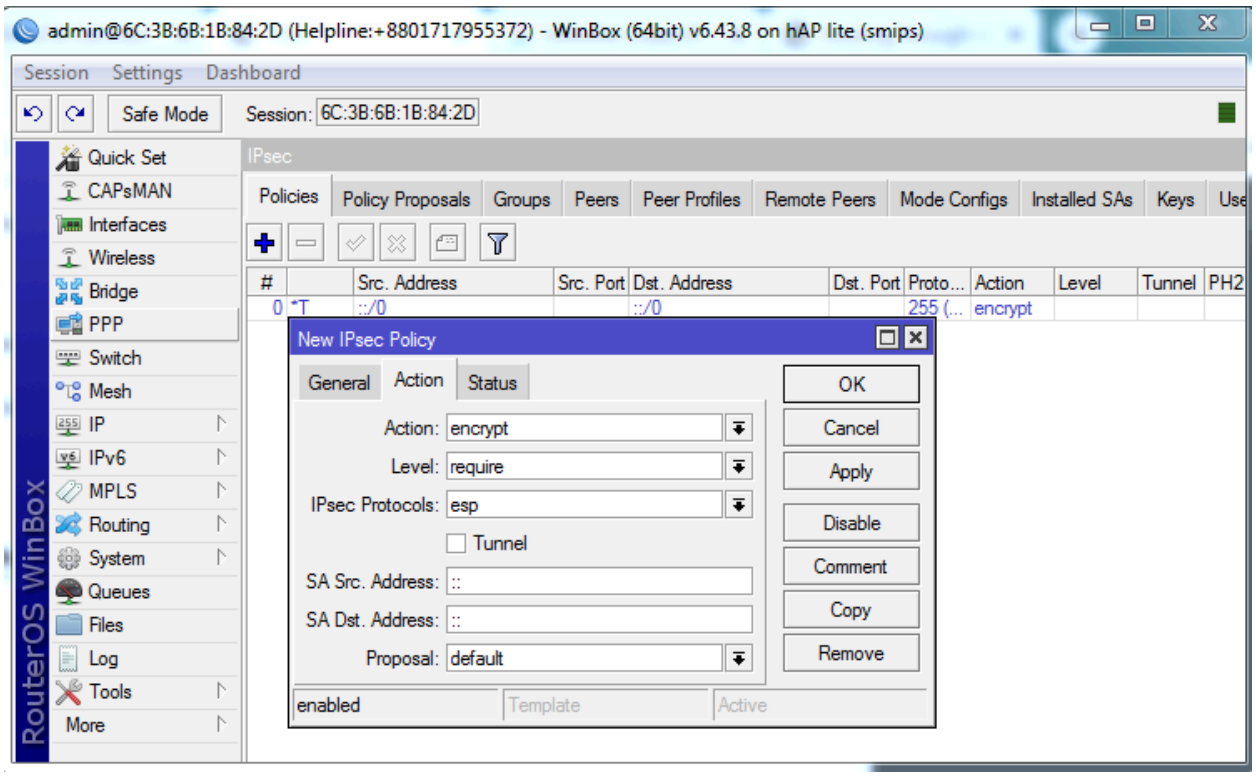
Flush SA



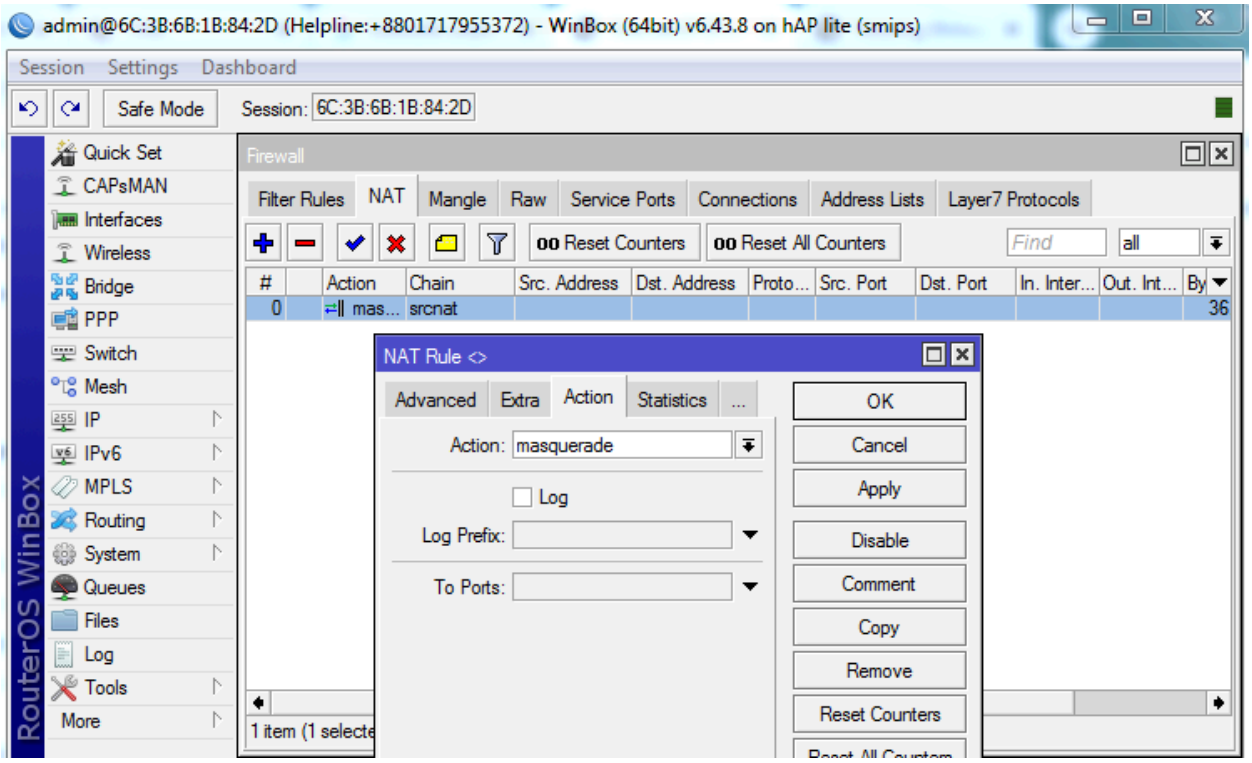
Create IPsec between two routers with NAT
Set peer



Set policy



Set NAT rules



Confirm the secure link is established
Mikrotik Bangla tutorial- 06(MTCNA) : How to tunnel configure by Mikrotik
Check on my youtube channel

HotSpot
Introduction (Concepts)
The MikroTik HotSpot Gateway provides authentication for clients before access to public networks .

HotSpot Gateway features:
1 different authentication methods of clients using local client database on the router, or remote RADIUS server;
2. users accounting in local database on the router, or on remote RADIUS server;
3. walled-garden system, access to some web pages without authorization;
4.login page modification, where you can put information about the company;
5. automatic and transparent change any IP address of a client to a valid address;

Usage environments
Campus, Hotel, cafe etc

Setup HotSpot with default settings
The simplest way to setup HotSpot server on a router is by /ip hotspot setup

HotSpot Login Methods

HTTP PAP - simplest method, which shows the HotSpot login page and expect to get the authentication info (i.e. username and password) in plain text.

HTTP CHAP - standard method, which includes CHAP challenge in the login page. The CHAP MD5 hash challenge is used together with the user's password for computing the string which will be sent to the HotSpot gateway. The hash result (as a password) together with username is sent over network to HotSpot service (so, password is never sent in plain text over IP network).

HTTPS - the same as HTTP PAP, but uses SSL protocol to encrypt transmissions. HotSpot user just sends his/her password without additional hashing (note that there is no need to worry about plain-text password exposure over the network, as the transmission itself is encrypted).

HTTP cookie - after each successful login, a cookie is sent to the web browser and the same cookie is added to active HTTP cookie list. Next time the same user will try to log in, web browser will send the saved HTTP cookie. This cookie will be compared with the one stored on the HotSpot gateway and only if source MAC address and randomly generated ID matches the ones stored on the gateway, user will be automatically logged in using the login information (username and password pair) was used when the cookie was first generated.

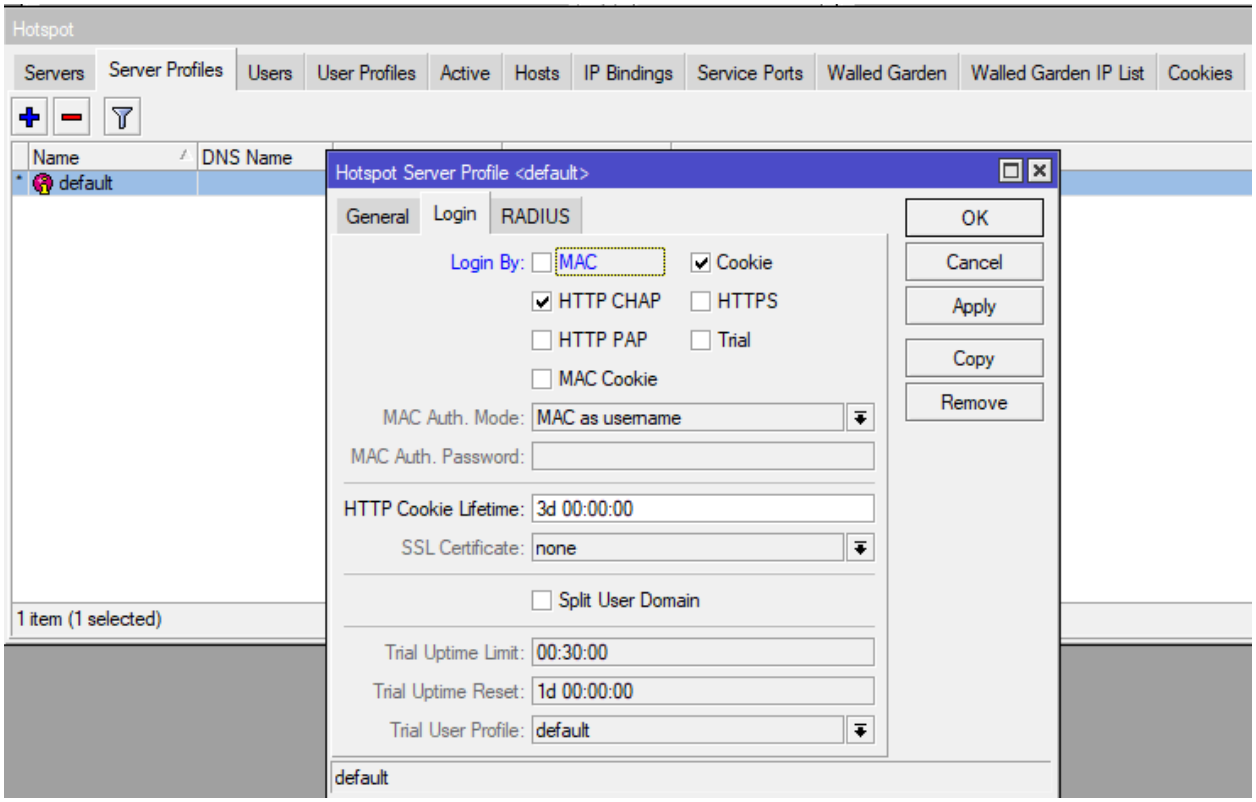
MAC address - try to authenticate clients as soon as they appear in the hosts list (i.e., as soon as they have sent any packet to the HotSpot server), using client's MAC address as username.

Trial - users may be allowed to use the service free of charge for some period of time for evaluation, and be required to authenticate only after this period is over.

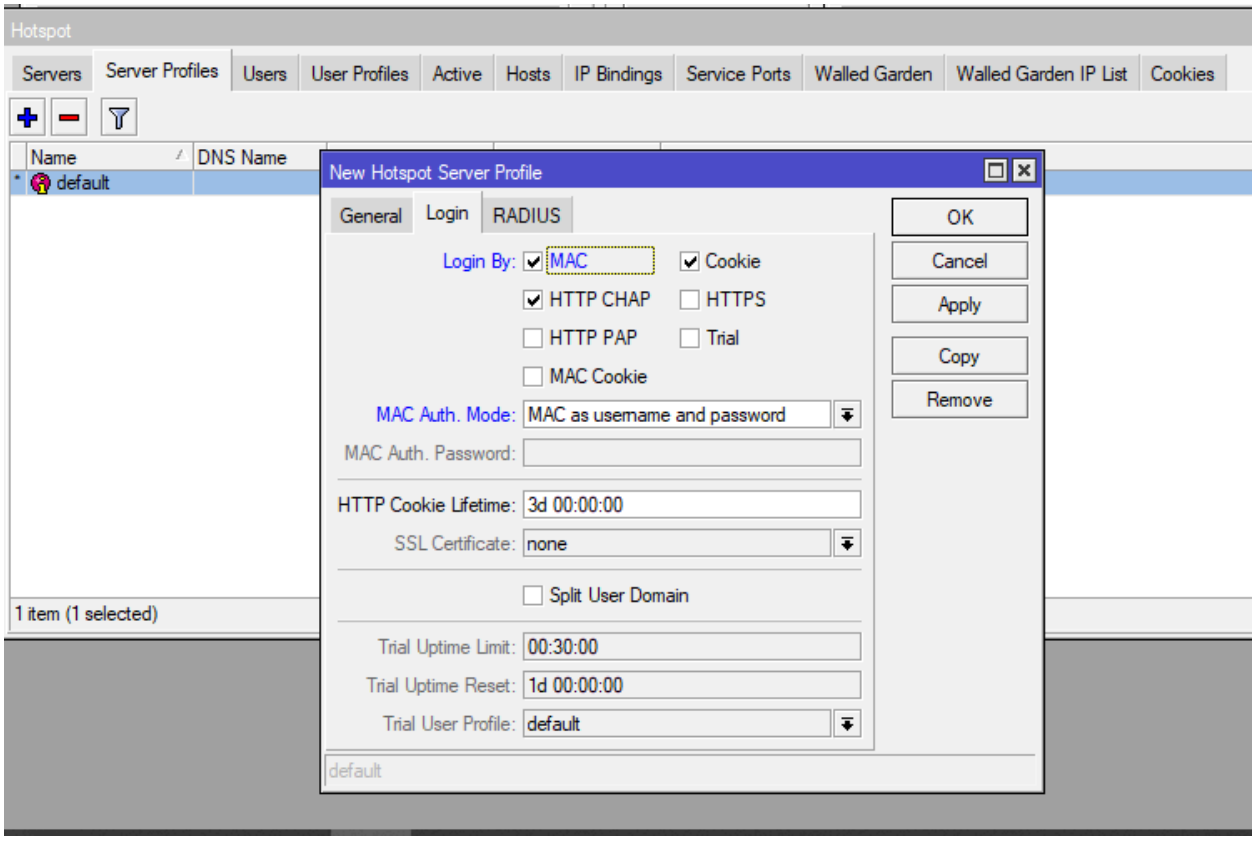
RADIUS: Remote Authentication Dial-In User Service (RADIUS) is a networking protocol, operating on ports 1812 and 1813, that provides centralized Authentication, Authorization, and Accounting (AAA or Triple A) management for users who connect and use a network service

MikroTik Hotspot Users
Add users
/ip hotspot user

Set MAC-address for user



Set MAC-address for username

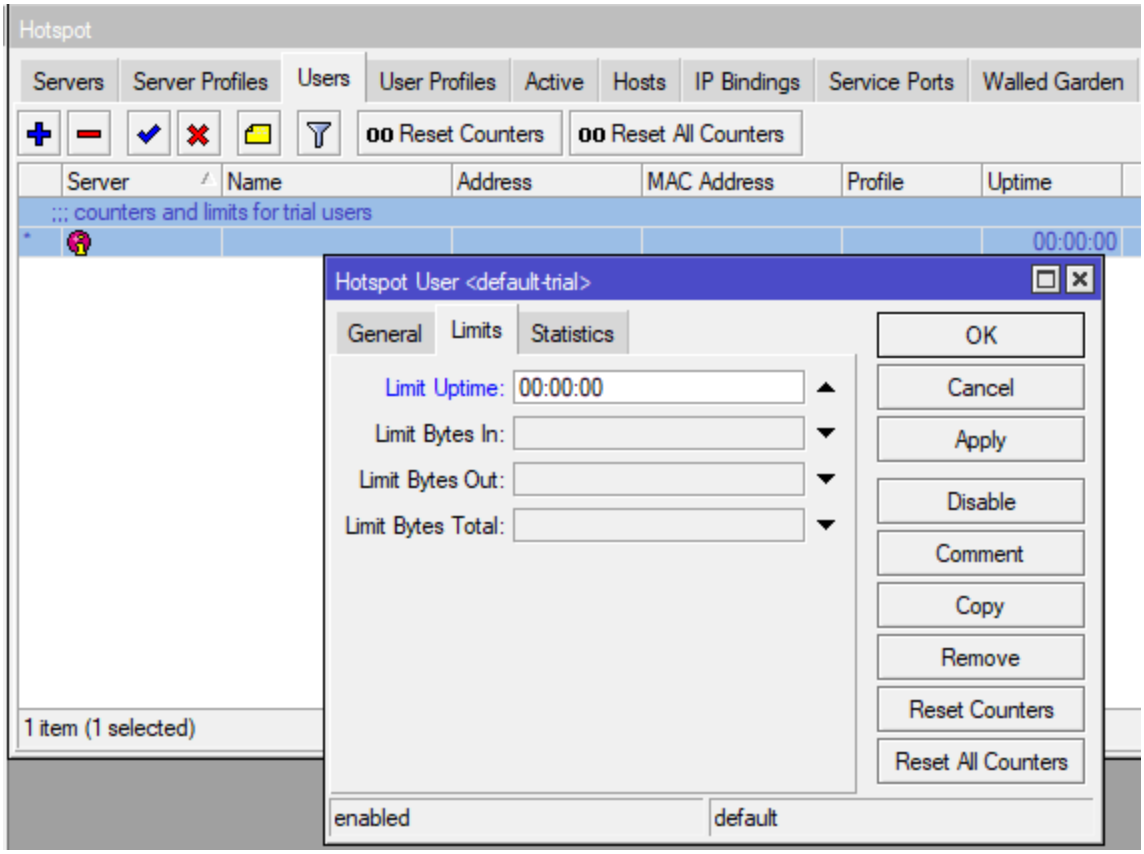


Limit Uptime and Limit Bytes In/Out

Limit Uptime
Uptime limit for the HotSpot client, user is disconnected from HotSpot as soon as uptime is reached.

Limit-bytes-in
Maximal amount of bytes that can be received from the user. User is disconnected from HotSpot after the limit is reached.

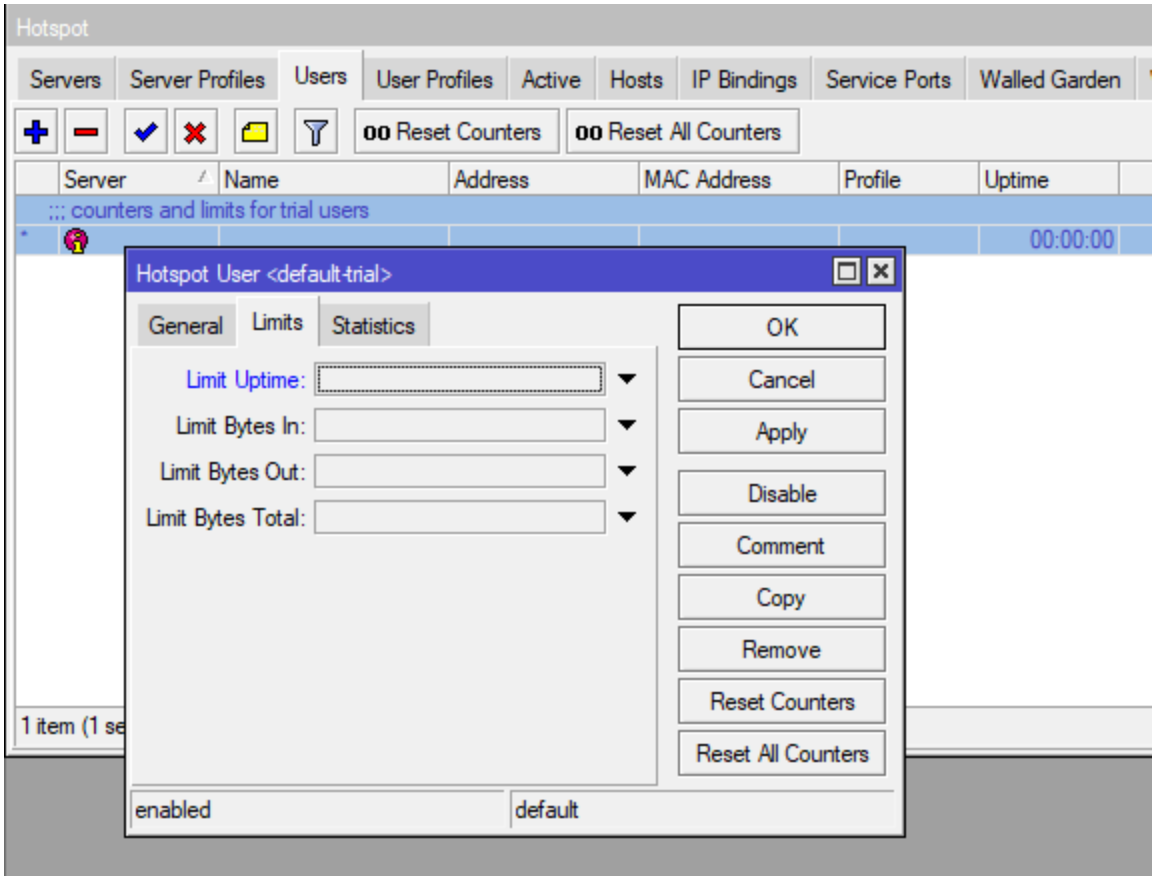
Limit-bytes-out
Maximal amount of bytes that can be transmitted from the user. User is disconnected from HotSpot after the limit is reached.



Reset limits for user

tool user-manager user reset-counters

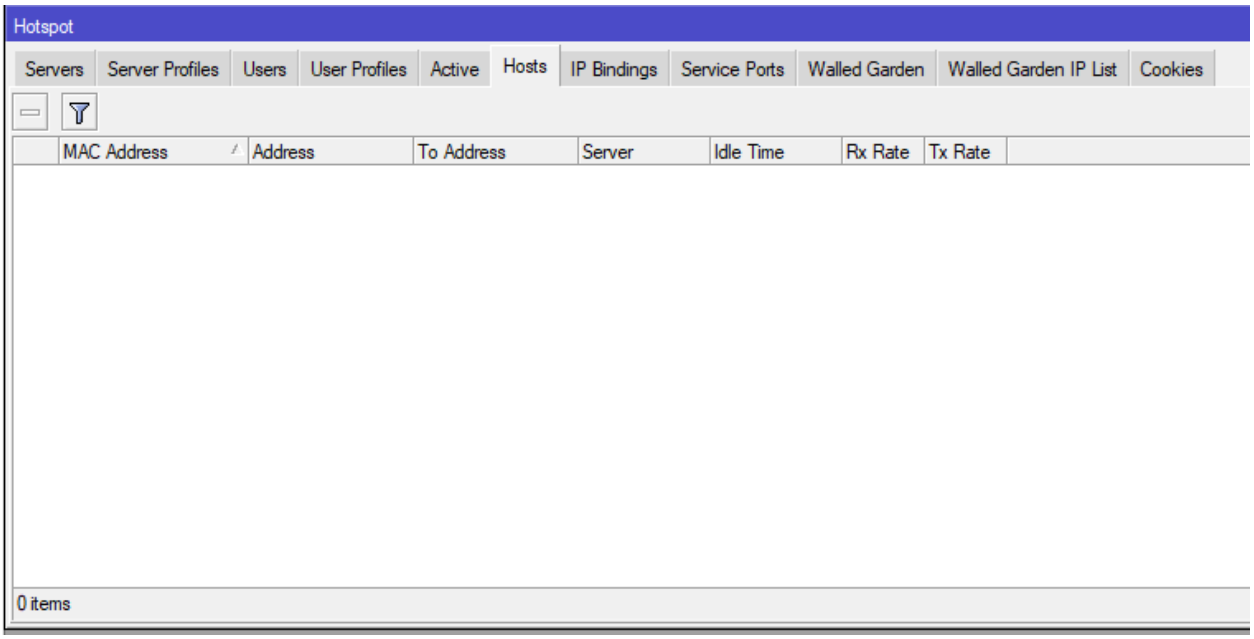
Or



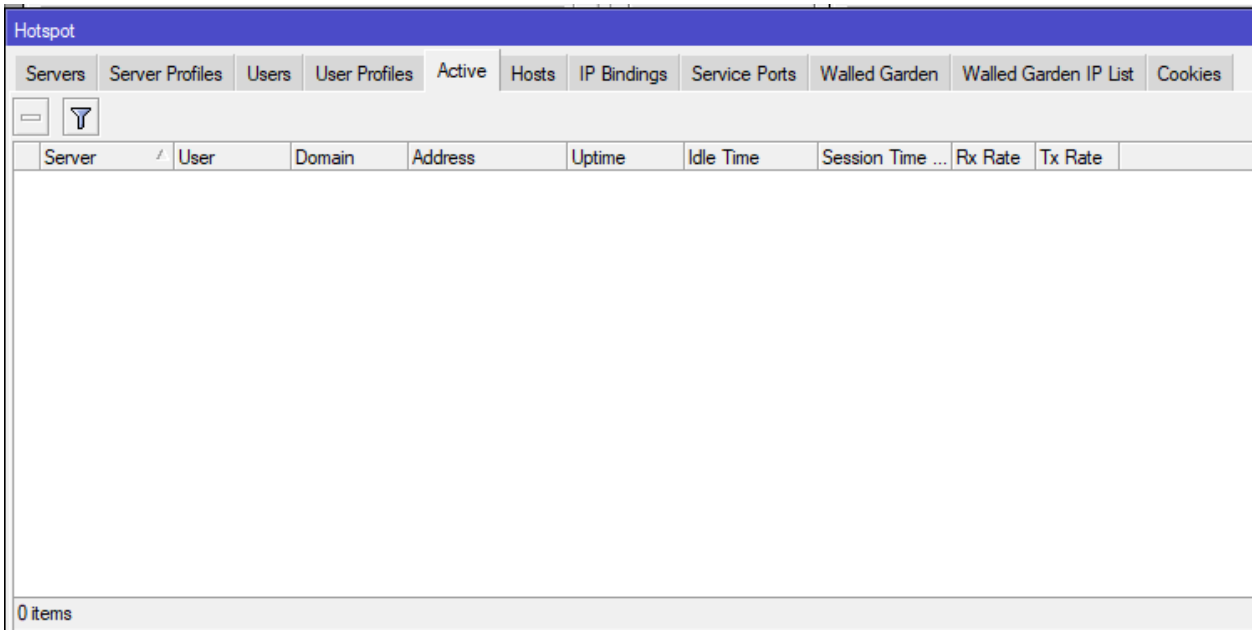
Monitor Users

Host Table

ip> hotspot> host



Active Table



SNMP for users

/ip hotspot active print count-only

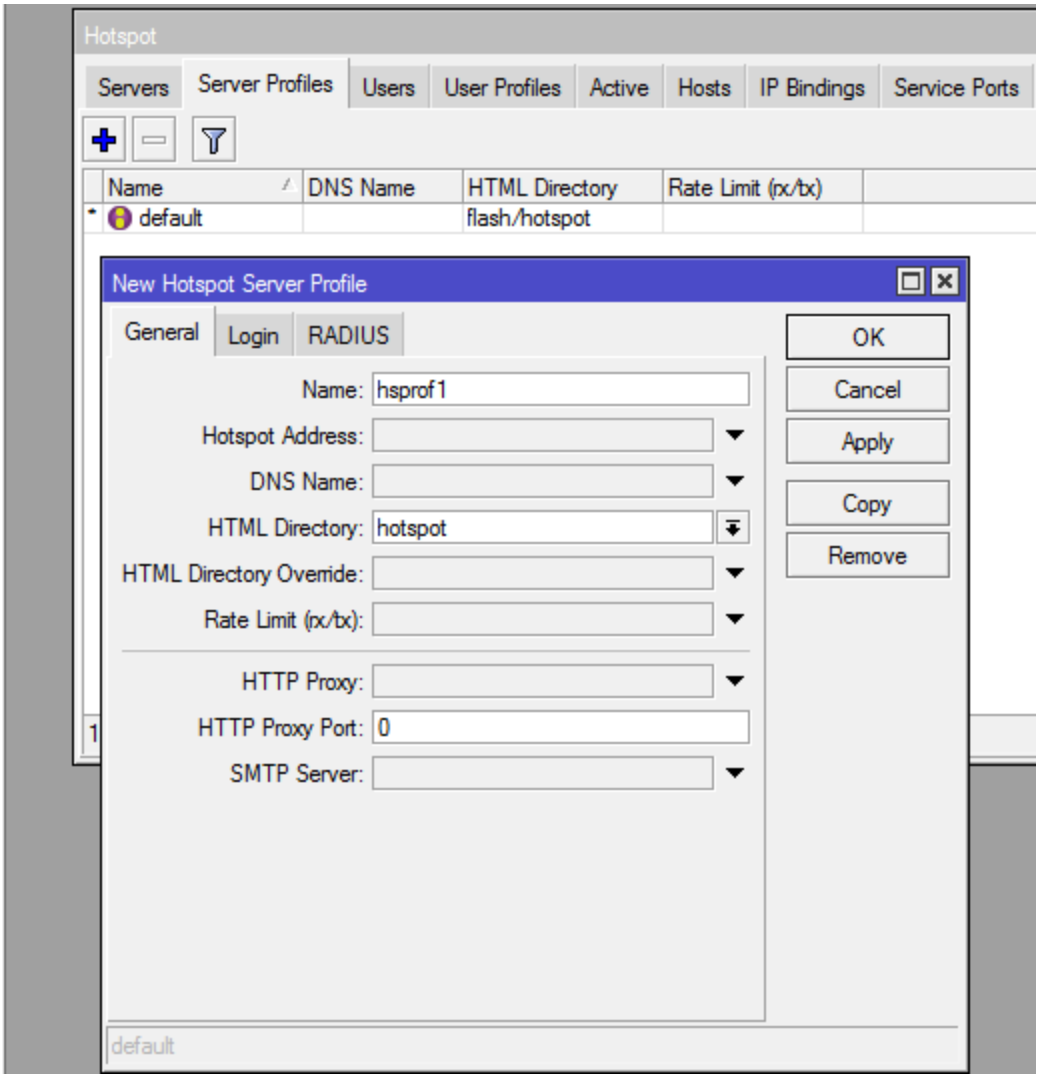
Or

Try over dude

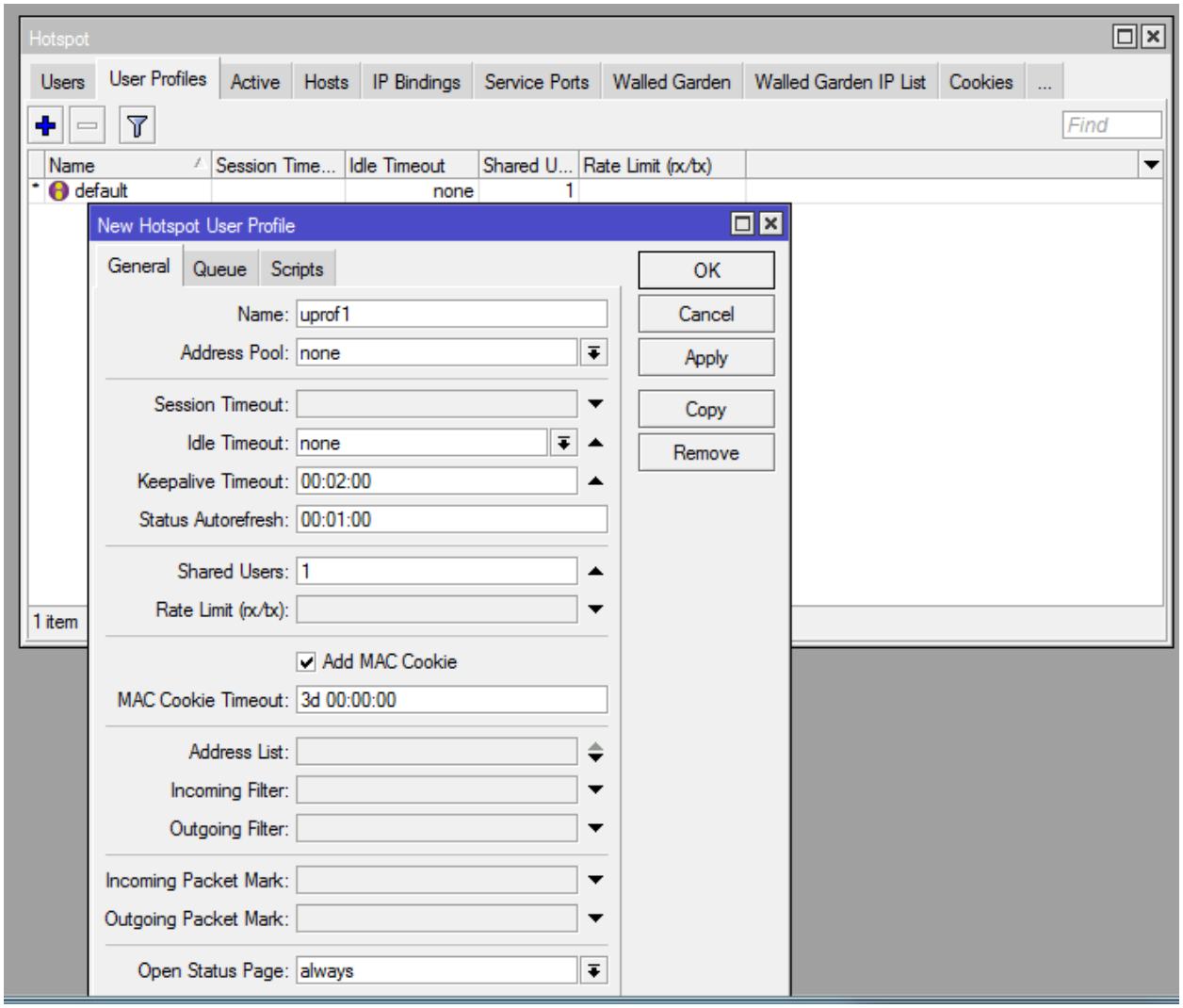
Profile

/ip hotspot profile

This submenu contains list of Hotspot server profiles. There may be various different HotSpot systems, defined as HotSpot Server Profiles, on the same gateway machine



Keepalive timeout

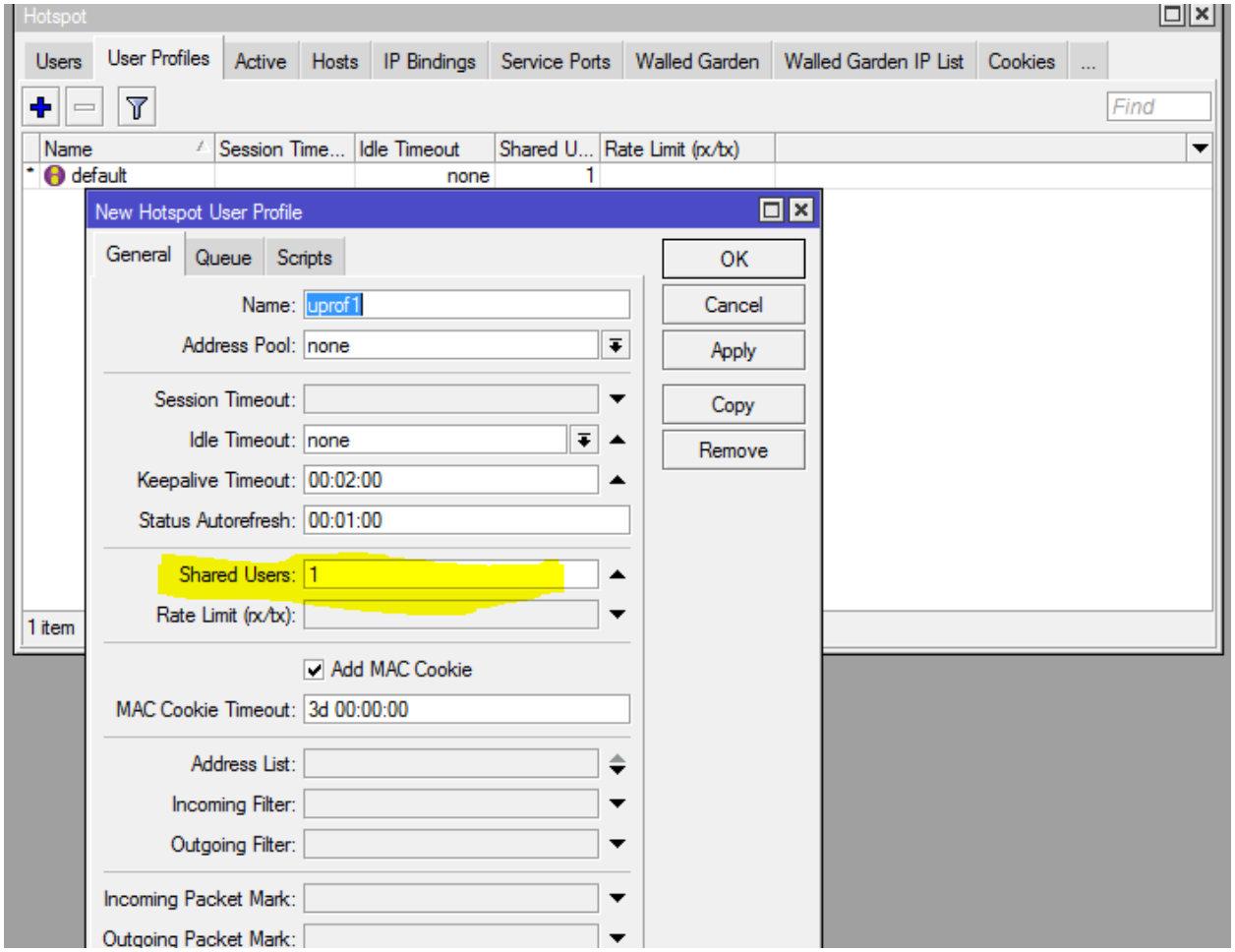


Shared users

Shared users = the number of times this same user account can connect simultaneously. Example -We allow the user to sign-in for their main computer and their phone only for basic service.

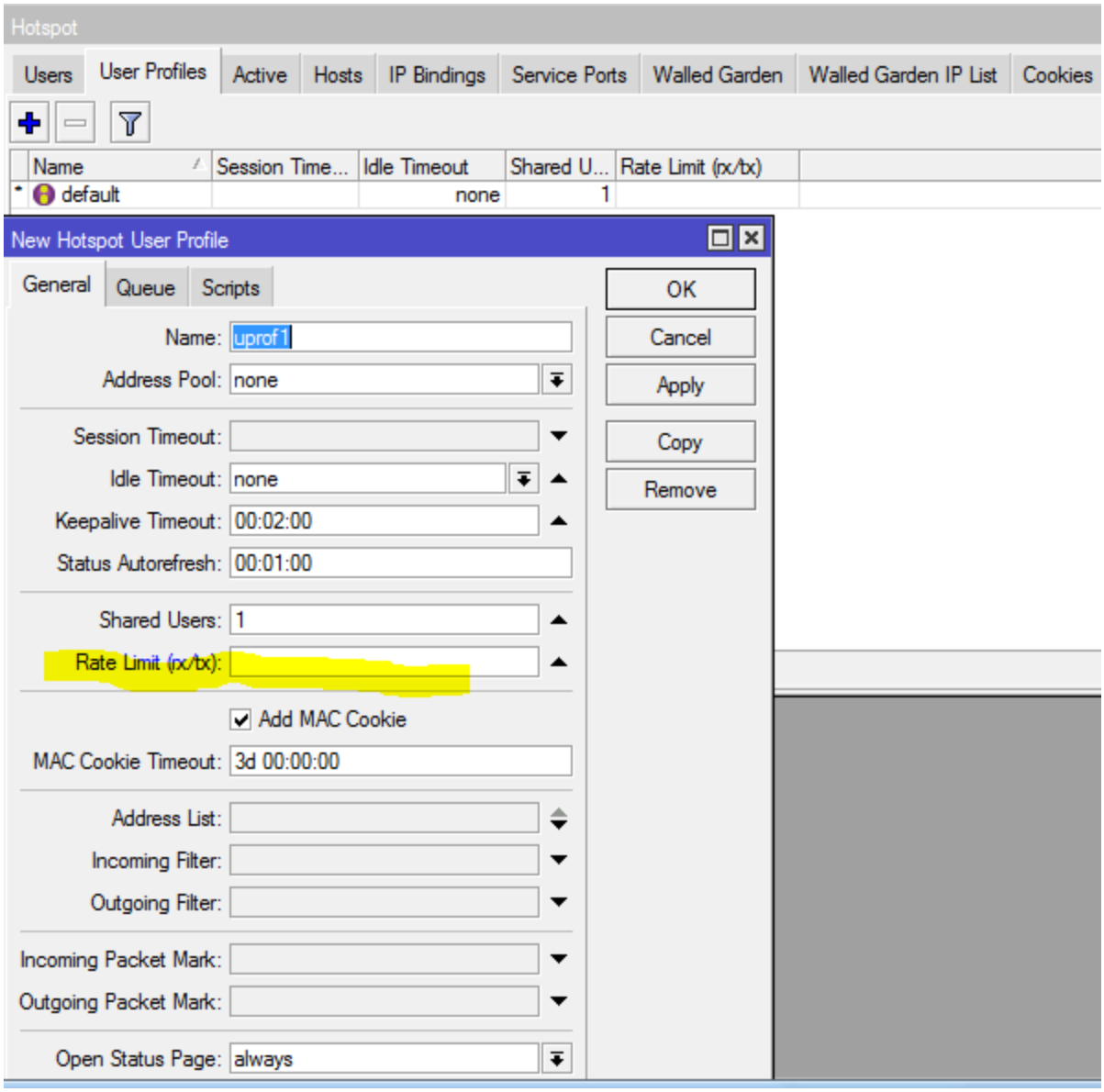
If they try more devices the log-in screen shows the "session limit reached" error.

IP->Hotspot->User Profiles tab for Shared Users count

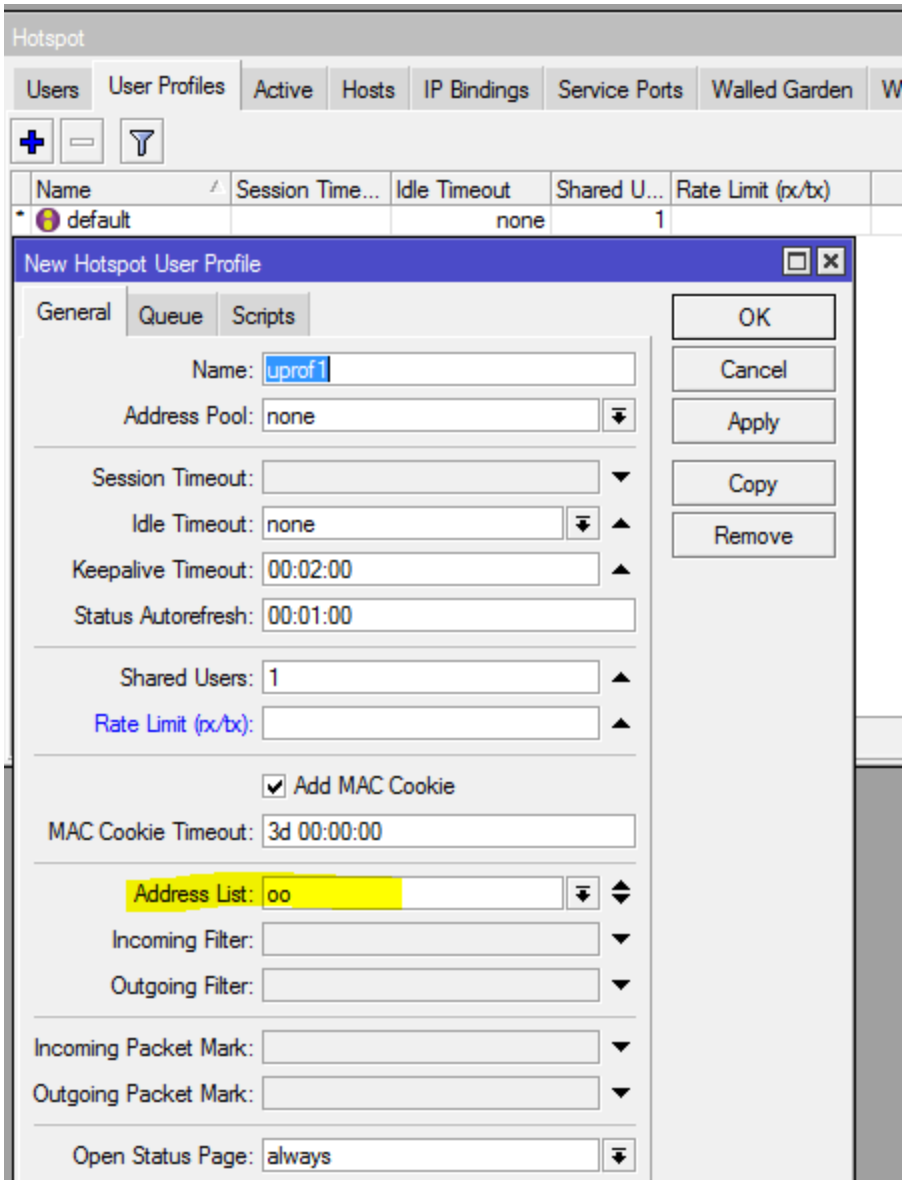


Rate-Limit
In "/ip hotspot user profile", you can set the rate-limit. That will create a simple queue for you.

For example, this is 256k max upload, and 512k max download (viewed from the client)

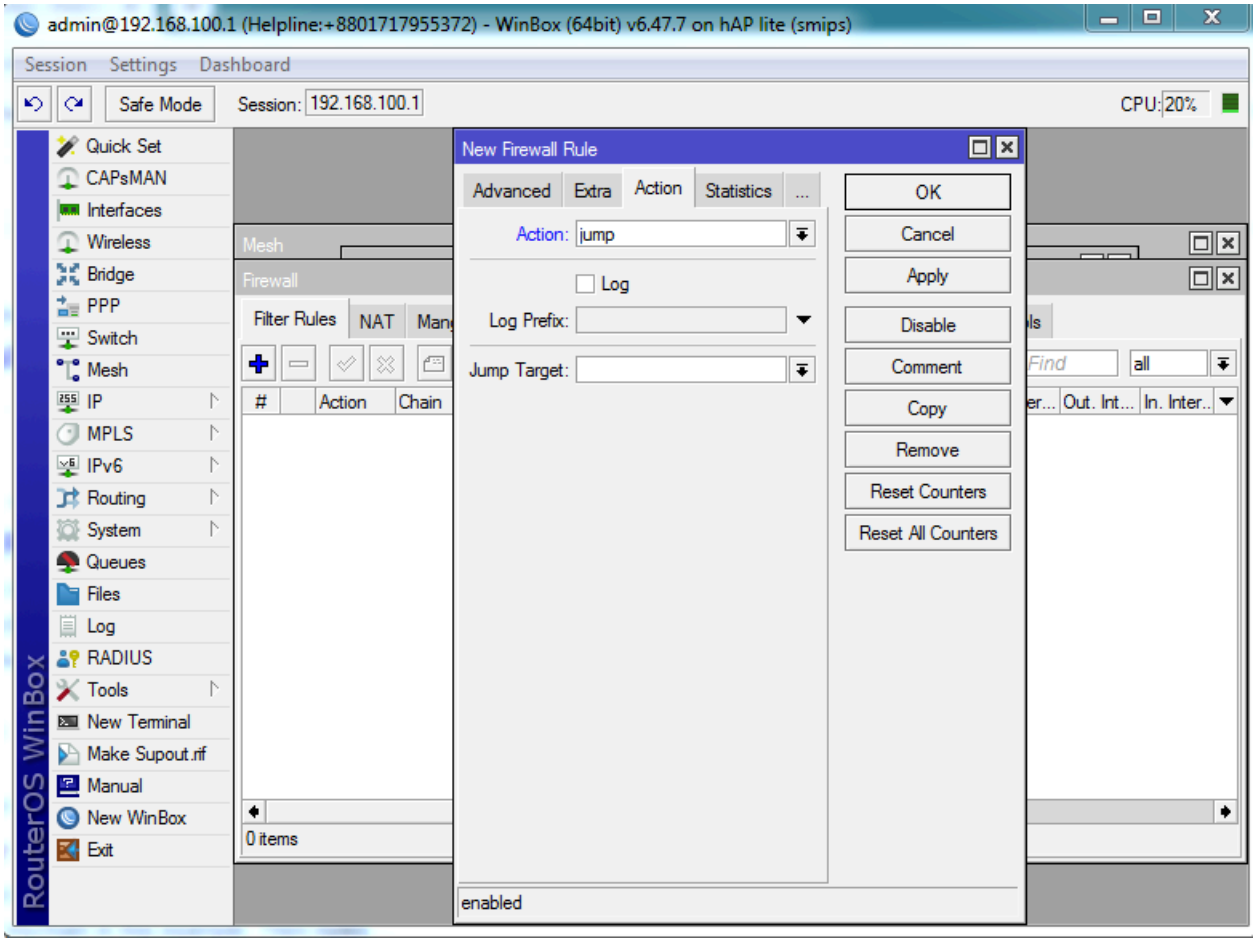


Address-list



Incoming/Outgoing filter

incoming-filter
Name of the firewall chain applied to incoming packets from the users of this profile, jump rule is required from built-in chain (input, forward, output) to chain=hotspot
outgoing-filter
Name of the firewall chain applied to outgoing packets from the users of this profile, jump rule is required from built-in chain (input, forward, output) to chain=hotspot



Incoming/Outgoing Packet Mark

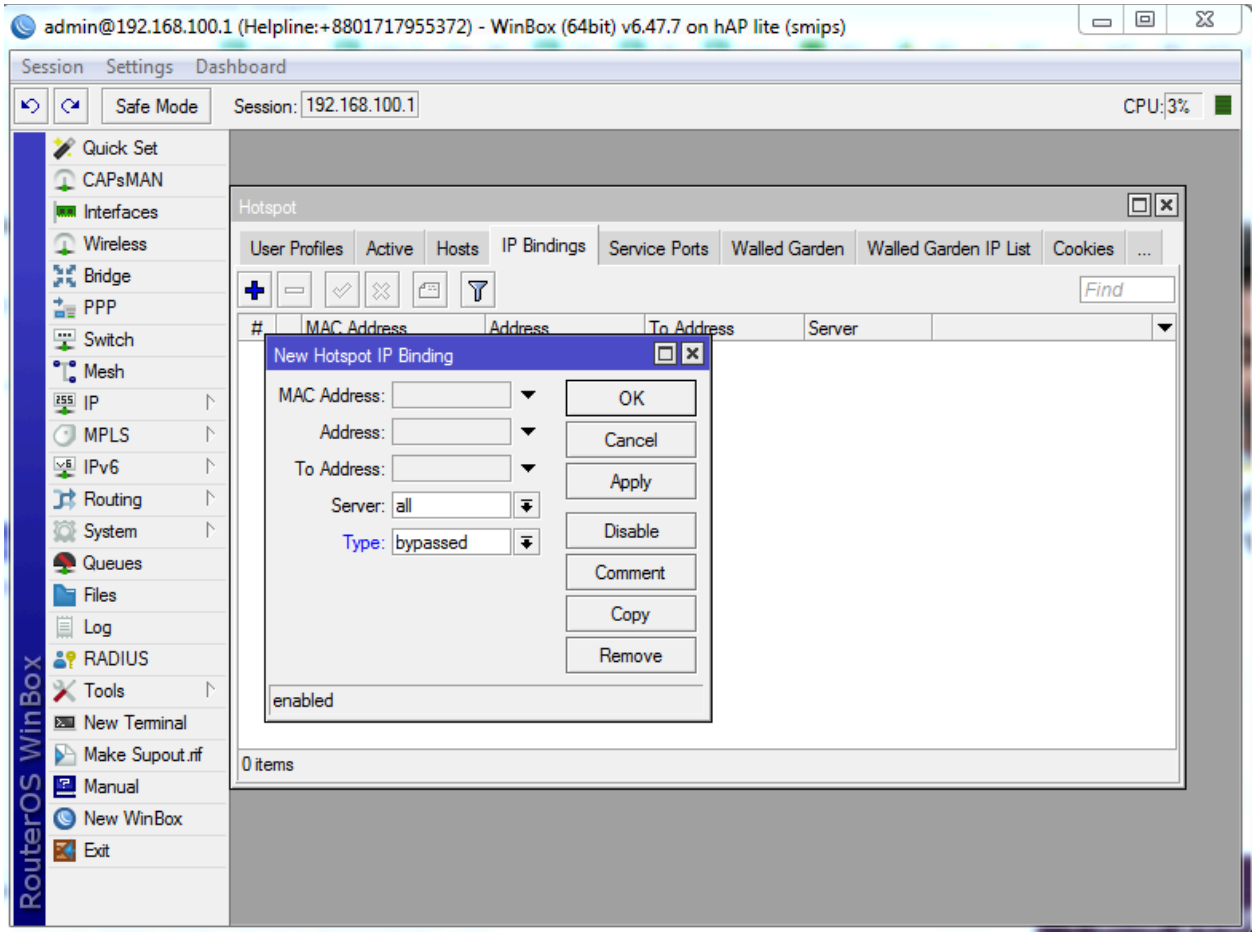
incoming-packet-mark
Packet mark put on incoming packets from every user of this profile

outgoing-packet-mark
Packet mark put on outgoing packets from every user of this profile

Bypass HotSpot

Mikrotik Hotspot System makes all Users must Login first in order to access the internet. Sometimes there are some users who are lazy to log in and want to be able to directly enter Hotspot without login or Bypass Login Hotspot Mikrotik. To accommodate this user request, we can take advantage of the Hotspot IP-Bindings feature.

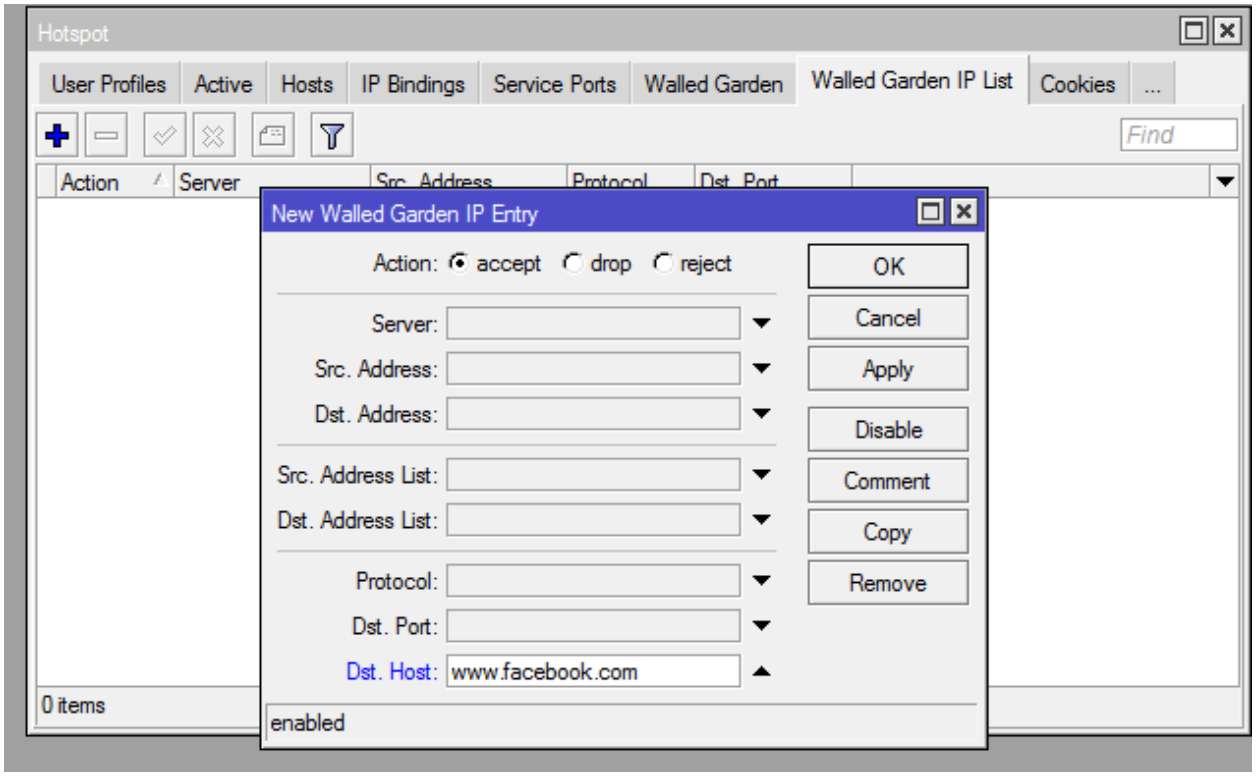
IP Bindings allows us to bypass certain user hotspots without having to go through the authentication process (login). In addition, IP Binding also allows us to block certain users so that he can not access the hotspot network.



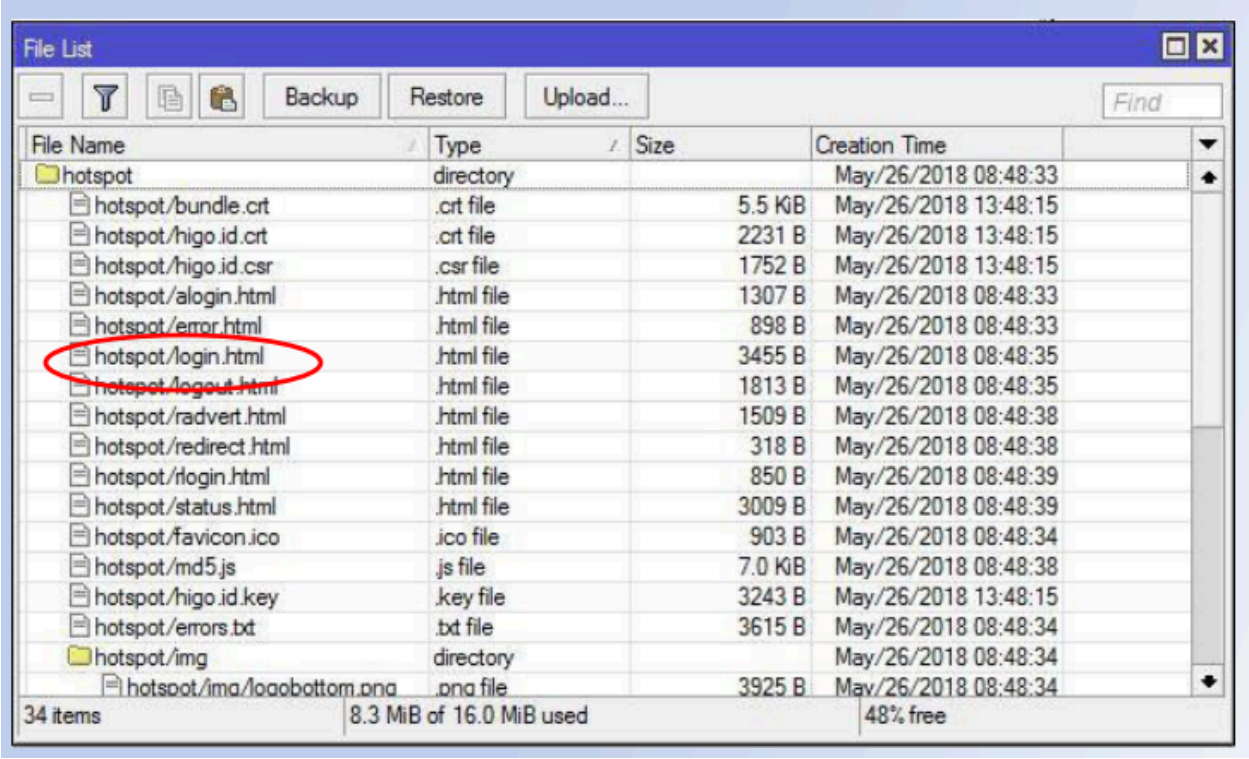
Walled Garden and Walled garden IP

/ip hotspot walled-garden

HTTP walled-garden menu permits authentication bypass settings for HTTP and HTTPs resources.



Customize HotSpot
Advertisement
Customize pages

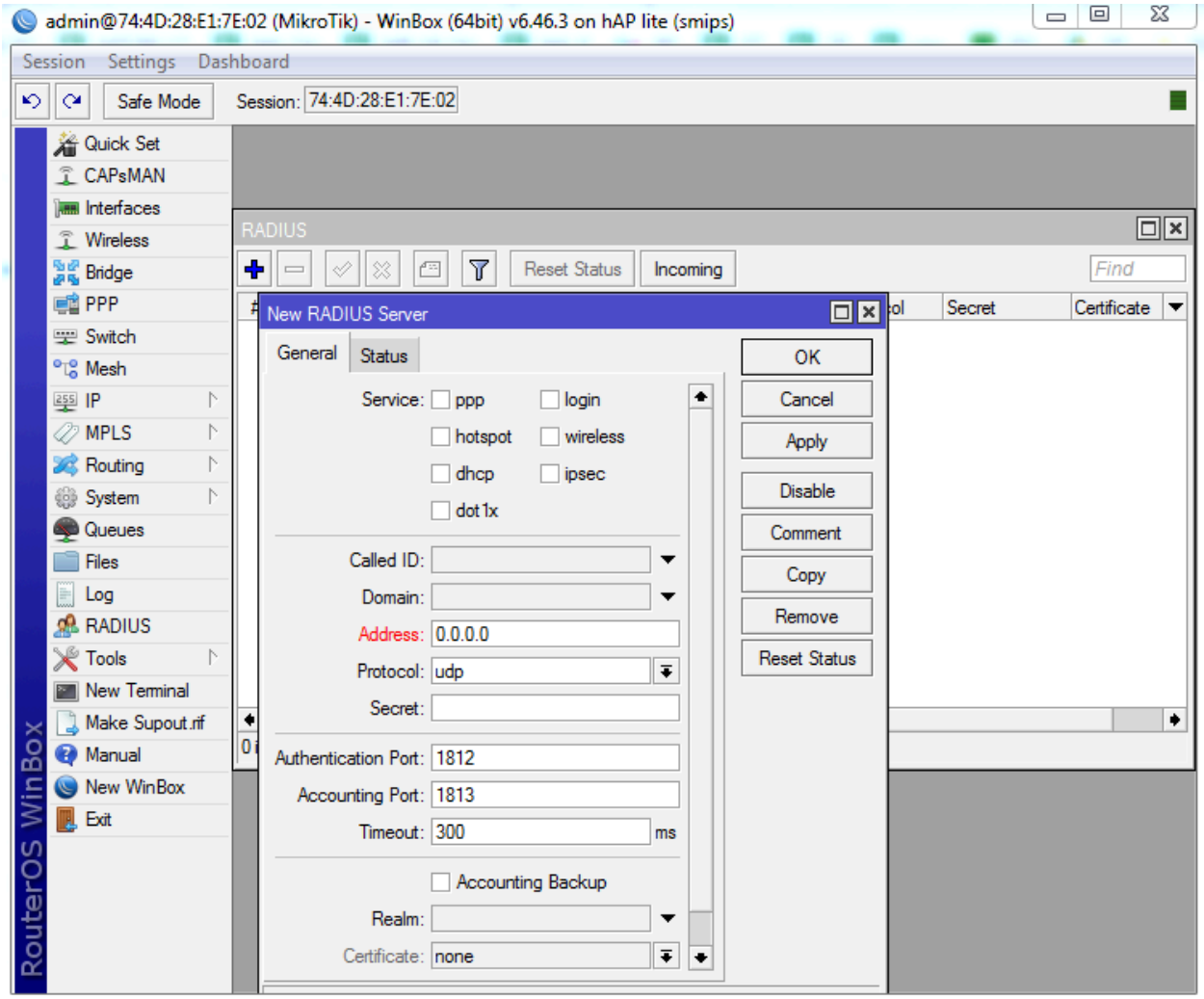


RADIUS client

RADIUS, short for Remote Authentication Dial-In User Service

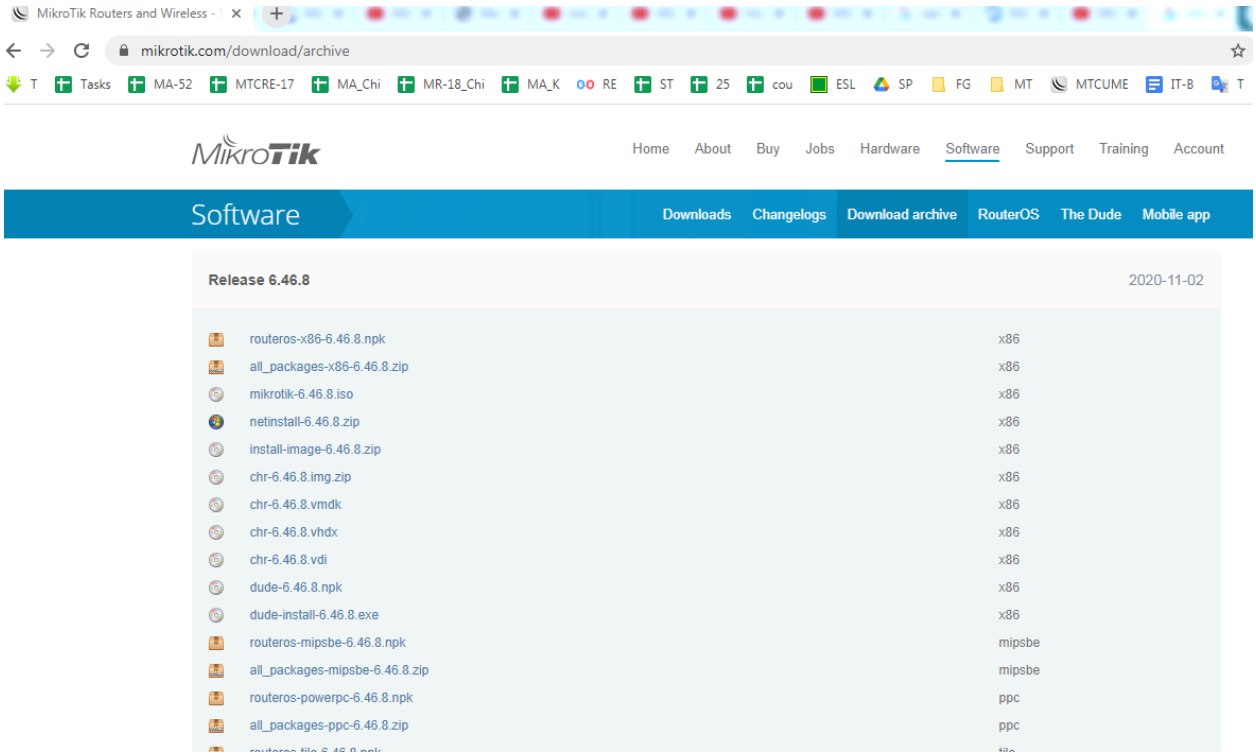
RADIUS client

- Add radius client
- Set service
- Use RADIUS for the specific service



RADIUS server

- User manager
- Install the latest user-manager
- Add routers
- Add users
- Set profile



RADIUS incoming

Radius --> Incoming --> Port

This is the UDP port that will be listening for CoA(Change of Authorisation) packets.